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ABSTRACT

In the midst of COVID-19 pandemic and technology advancement, e-wallet has become one of the common payment method in the market. This research aims to examine the influence of the factors on the intention of using e-wallet. The factors that will be discussed in this study is perceived usefulness, perceived ease of use, perceived security, subjective norm and attitude. In this study, the primary data will be collected through questionnaire using the online survey portal – Google Form. The sampling method and technique for this study is convenience sampling of non-probability sampling method. Total 245 responds have been collected, in which the survey is conducted within Malaysia. The result of the study revealed that perceived usefulness, perceived security and attitude has significant influence on the intention to use e-wallet. On the other hand, the effect of perceived ease of use and subjective norm have on the intention of adopting e-wallet is insignificant.

Keywords: E-wallet, Covid-19 pandemic, Intention to use

CHAPTER 1

1.0 Introduction

1.1 Introduction of the Chapter

As a result of Industry Revolution 4.0, the world has developed to become automated, greatly utilised artificial intelligence and robotics technologies. All these innovations have change the way people live, work, and interact. In spite of this, nowadays, people can use their mobile phones as virtual bank cards, perform internet banking transactions, check balance from mobile phone remotely without using automated teller machines (ATMs) (Lim et al., 2019). This is one of the prevalent change in today's society, the innovation of E-Wallet.

E-wallet is the acronym of Electronic Wallet, in which it is also known as digital wallet. It is an app-based payment system that allow users to make payment, receive, transfer or top-up funds via mobile devices, replacing the physical wallet (Andrew et al., 2019). As the e-wallet system can be integrated with several bank accounts, credit and debit cards, it provide convenience to the users since users can make their payment with their fingertips instead of physically swiping off the cards (Abdullah et al., 2020). Another benefit that users can gain from using e-wallet is security risk can be reduced as e-wallet eliminate the need for users to bring physical wallet and it allow users to store their cards in the system.

Nevertheless, new risk arise when using e-wallet, which is threat of cyber security. For instance, hacking, phishing attacks and malware. Besides that, e-wallet relies on the availability of internet connection, therefore, it would be inconvenience if there is no internet coverage or the phone is running out of battery (Abdullah et al., 2020).

1.2 Research Background

In the recent years, the development of cashless society is the sole goal for many countries around the world where people make transaction without using physical cash. Instead, cards and electronic methods are used to perform financial transactions (Rahman et al., 2021; Abdullah et al., 2020). According to World Payment 2020, the volumes of global cashless transaction have increased by 14% and amount to 708.5 billion during 2018 to 2019. This is the highest growth rate throughout the past decade (Rahman et al., 2021). According to Dataportal 2022, the global e-wallet transactions in 2022 have increased by 24.5% to 6.75 trillion dollar (figure 1).

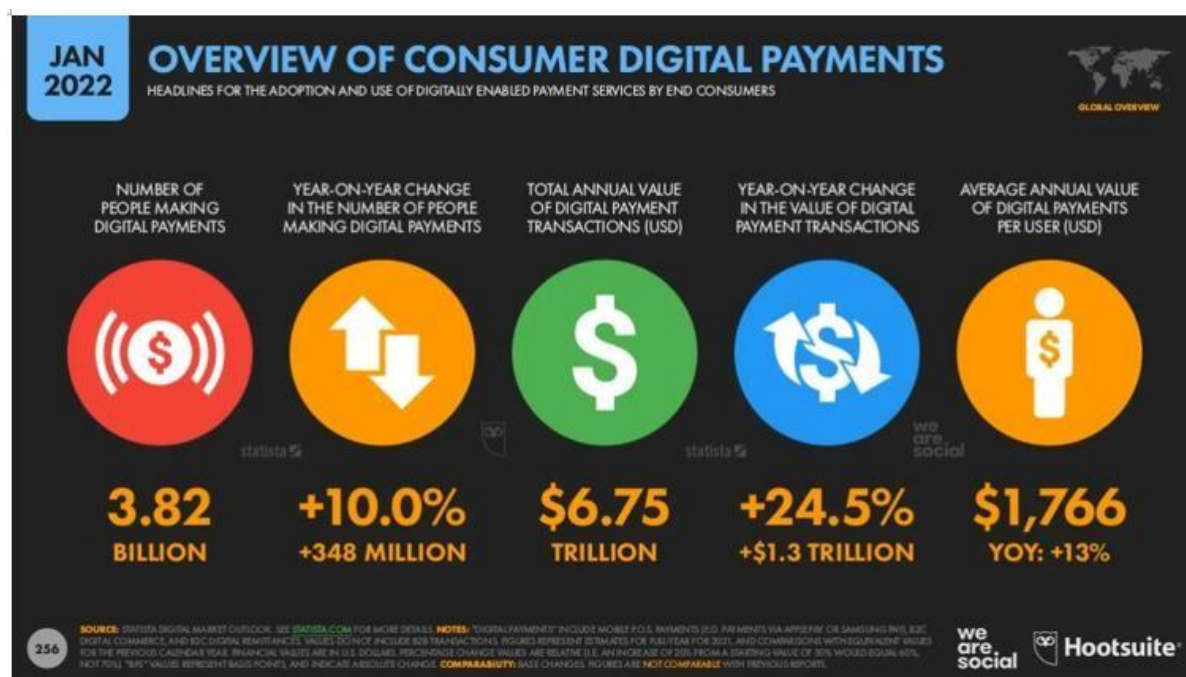


Figure 1.1 Digital payment growth

(source: Dataportal, 2022)

COVID-19 outbreak has led to rise in digital wallet adoption and usage in 2020 (GlobeNewswire, 2021). During this health crisis, World Health Organization (WHO) has encouraged people to follow the physical distancing policy and do contactless activities, including making payment transactions. WHO recommended people to use electronic money if possible because the virus can be transmitted through the physical money which expose people in danger if he or she touched the physical money (Aji et al., 2020). As a result, the adoption of e-wallet and cashless transactions in Malaysia have seen an upsurge during pandemic outbreak due to Movement Control Order (MCO) (Birruntha, 2020). In Malaysia, the popular e-wallet applications are Touch 'n Go E-wallet, Grab Pay, Boost Pay and Fave (Aji et al., 2020).

This pandemic has also changed the consumers buying behaviour where many consumers have shift to online shopping from brick-and-mortar channels and eventually trigger the need for e-wallet as a medium of payment. It is shown in figure 2, where the e-commerce market in Malaysia has grown up by 37.3% in during 2021. As e-commerce grows, it also drive the growth of e-wallet payment in Malaysia. This is evident in figure 3 where Malaysia's digital payment has increased by 21.7% (Dataportal, 2021).



Figure 1.2 E-commerce growth in Malaysia

(source: Dataportal, 2021)

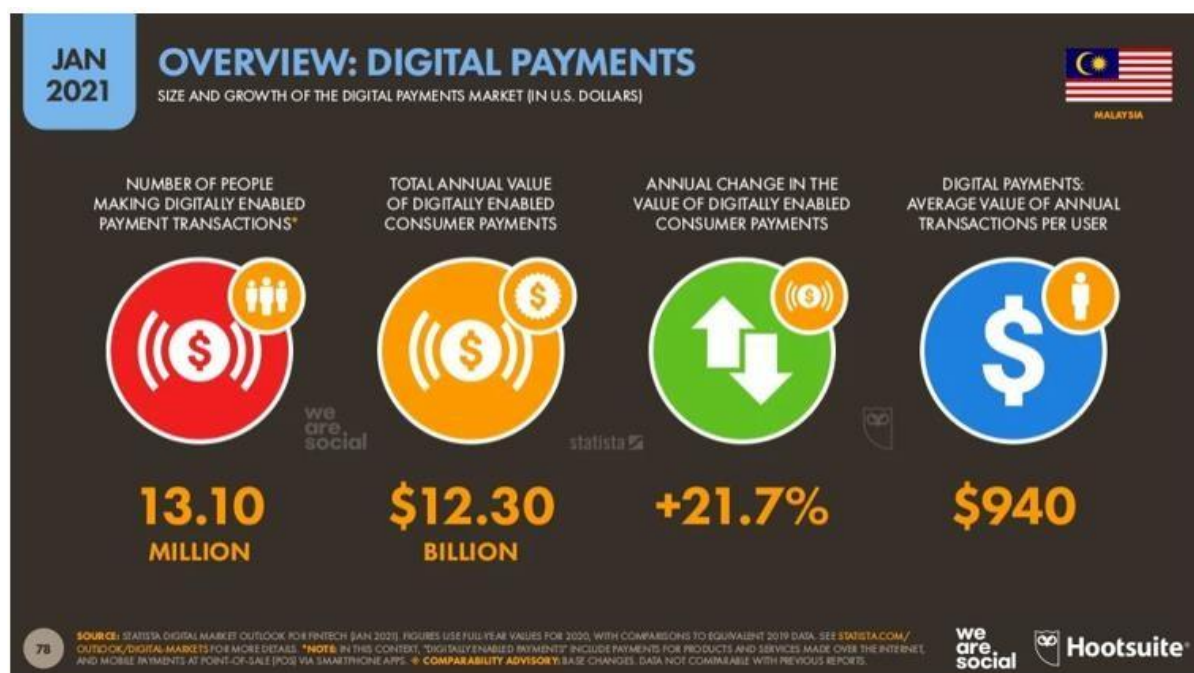


Figure 1.3 Digital payments in Malaysia

(source: Dataportal, 2021)

1.3 Problem Statement

With the technology advancement and influence of COVID-19 health issue, e-wallet has become an easier and faster payment method as it also provide various benefits and functions such as bill payment and online transactions (Wong & Mohamed, 2021). Nowadays, the whole world is being digitalized and payments are made through mobile phones have replaced primitive payment methods as the most convenient way to settle payments. Technology has made it necessary for consumers to rely on their mobile phones because it is faster, easier and more convenient to perform day-to-day transactions on their mobile phones. Additionally, e-wallet offers the users with incentives such as reward points and cash back, to attract more users to pay with e-wallet. This mode of payment has already gained popularity in developed countries like United States and Europe because of its convenience and safety. Furthermore, in developing countries such as China and India, many people tend to use e-wallet as they do not visit banks and ATMs often. For instance, almost 94% of the payments in China are made using e-wallets like Alipay and Wechat. Similarly in India, more than 220 million of people use PAYTM, which is a digital payment app to make their payments (Nag & Gilitwala, 2019).

To in line with the global trend, Malaysia's government has implemented several initiatives such as ePENJANA e-wallet stimulus, to motivate more citizens in adopting the e-wallet (Aji et al., 2020). However, research showed that although e-wallet has been implemented for several years in Malaysia, the level of acceptance of e-wallet is still low provided the fact that this technology has already existing in Malaysia for some years (Abdullah et al., 2020). According to Andrew et al. (2019), despite the e-wallet phenomena is getting more apparent in Malaysia, but the adoption growth rate is still lower than other regional peers, for example, Singapore and China.

Datareportal (2022) stated that Malaysia has a lower adoption rate of mobile payment services with 23.2% as compared to the worldwide average rate of 25.8% (figure 4). Hence, the purpose of this study is to investigate what are the factors that actually influence the consumers' intention of adopting e-wallet.

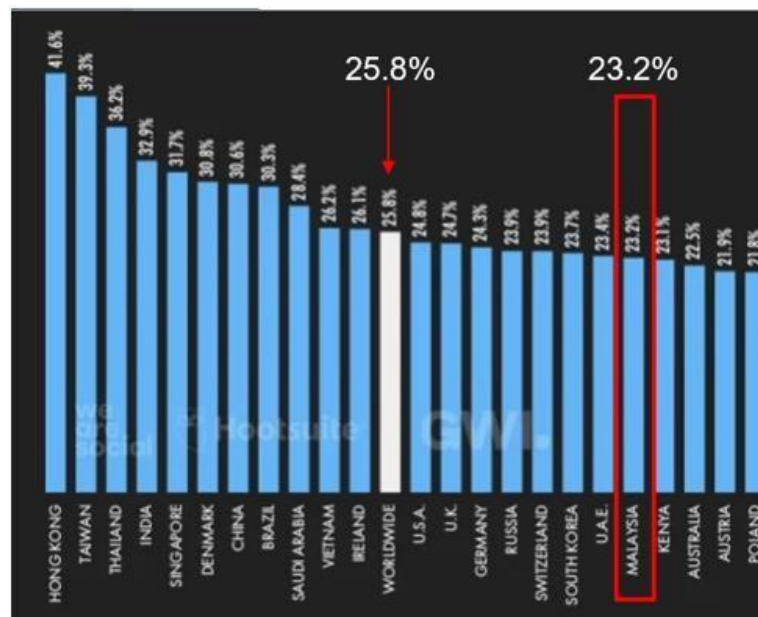


Figure 1.4 Mobile payment services user rate

(source: Dataportal, 2022)

1.4 Research Gap Analysis

During year 2020, a global pandemic, COVID-19 has started to appear in Malaysia and the local cases increased drastically day-by-day. In March 2020, the Malaysian government has enforced Movement under Control Order (MCO) where people are advised to practise social distancing and avoid contact between individuals (Birruntha, 2020). Because of the fear of coronavirus people, are now more aware of the need for electronic payments and e-wallets to prevent virus infection (Bavel et al., 2020). This has reiterate the importance of studying this topic to examine consumers' intention of adopting e-wallet. In Malaysia, e-wallet has been implemented for some years. However, the acceptance level of e-wallet among consumers is still quite low (Osman & Leong, 2021). Hence, there is a need to conduct a study to determine the factors that affect the user's acceptance of e-wallet.

Nevertheless, there are limited studies that focus on examining the influencing factors of consumers' intention to adopt e-wallet after experiencing the onset of COVID-19 pandemic (Daragmeh et al., 2021). In fact, in other developing countries such as China, India, Indonesia and Vietnam, a number of studies towards consumers' intention to use e-wallet had been conducted. Whilst, there are still limited findings about the influencing factors of e-wallet adoption by consumers in Malaysia (Osman & Leong, 2021). Therefore, this created a gap for further investigation. Furthermore, from the past studies, there is little literacy work focus on

how perceived security influence the consumers' intention of using e-wallet (Daragmeh et al., 2021; Abdullah et al., 2020; Andrew et al., 2019).

Hence, this study is proposed to fill the research gap in the past literature relating to the factors that influence the intention of adopting e-wallet in Malaysia after the onset of pandemic as there is not thorough literacy work has been done to this particular issue.

1.5 Research Question

1. What are the factors that influence the consumers' intention to use e-wallet?

1.6 Research Objectives

The objectives of this study are as below:

1. To investigate the relationship between perceived usefulness and consumers' intention to use e-wallet.
2. To determine the relationship between perceived ease of use and consumers' intention to use e-wallet.
3. To measure the relationship between perceived security and consumers' intention to use e-wallet.
4. To analyse the impact of subjective norm on consumers' intention to adopt e-wallet.
5. To examine the effect of attitude on consumers' intention to adopt e-wallet.

1.7 Significance of the Study

The findings from this study is to contribute for government regarding consumers' intention in adopting e-wallet and how to increase the adoption rate in Malaysia. Malaysia's policy makers also can uses this valuable findings to formulate relevant laws and regulations to accelerate the e-wallet adoption rate and strengthen the security on e-wallet so as to ensure the consumers can use e-wallet without fear of cyber-security risk.

In addition, the result of this study may help e-wallet providers to improve the systems and services in order to keep their users satisfied. E-wallets could be adopted widely in Malaysia when more users started to feel satisfied with the e-wallet. As a result, e-wallet penetration rate will be increased to establish a cashless society (Alam et al. 2021; Singh et al. 2020).

E-wallet is also being useful for frequent online shoppers. As online shopping has become a new trend in the market due to pandemic, e-commerce business is booming. Some e-commerce retailers have their own e-wallet system, for example, Shopee has their own e-wallet system which is known as ShopeePay, where customers are allow to pay directly through ShopeePay when checkout their purchases. To enhance the online shopping experience, e-commerce retailers are able to retrieve information from the findings of this study to improve or design an e-wallet payment system which linked to the online shopping platform so as to increase the efficiency and enhance customer convenience and satisfaction.

1.8 Summary

The first chapter has discussed the background of the issue regarding the low adoption rate of e-wallet in Malaysia. In the next chapter, past literatures and researches will be explained further to develop the theoretical framework.

CHAPTER 2

2.0 Literature Review

2.1 Introduction of the Chapter

The second chapter will be studying on the past researches on similar topic and the past literatures that are in line with the research objectives (discussed in chapter 1). The three main aspects that will be discussed in this chapter are variables (both dependent and independent), hypotheses development and theoretical framework development.

2.2 Theoretical Background

2.2.1 Technology Acceptance Model (TAM)

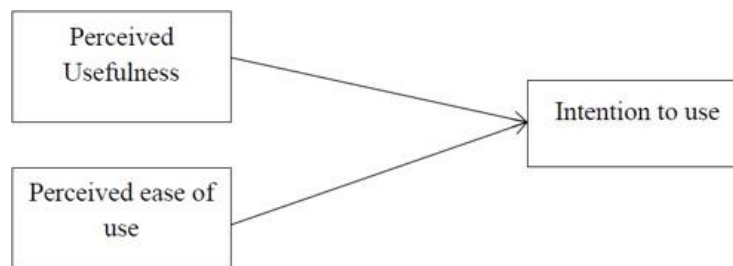


Figure 2.1 TAM Research Model (Davis, 1989)

Technology Acceptance Model (TAM) is a theory proposed by Davis (1989), which it is known as the most influential and commonly used theory in explaining the individual's acceptance of technology and system. It aims to identify the factors that influence in individual's intention to accept a new technology. This model assumes that the acceptance level of an individual towards a technology or system is influenced by two factors, which they are perceived usefulness (PU) and perceived ease of use (PE) (Lee et al., 2003). These two variables are often acting as the determinants to affect individual to accept to use the e-wallet.

Perceived usefulness means the confidence of an individual in which he or she believe that the particular technology and system will help them to improve their work. Perceived ease of use refers to the degree of individual to believe that the technology and system will not be complex to function (Davis, 1989).

2.2.2 Unified Theory of Acceptance and Use of Technology (UTAUT)

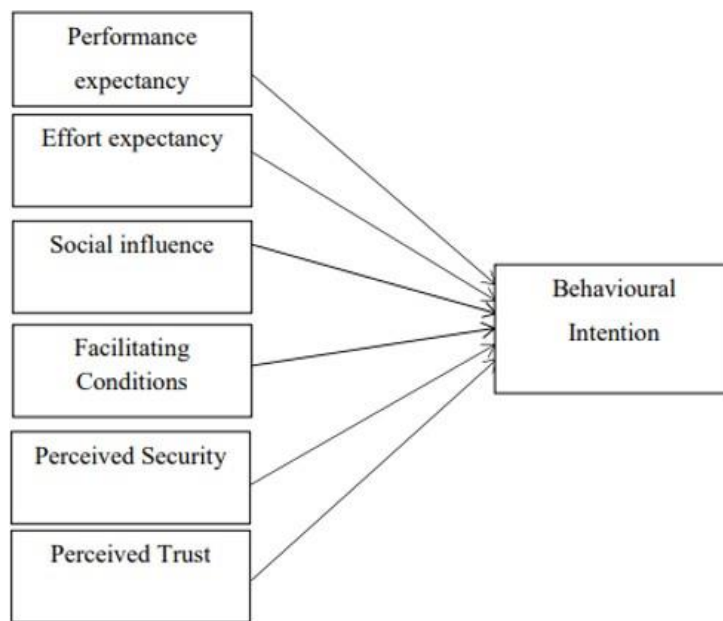


Figure 2.2 UTAUT Research Model (Venkatesh et al., 2003)

Unified Theory of Acceptance and Use of Technology (UTAUT) was created by Venkatesh et al. (2003). It was developed as an extension of Technology Acceptance Model (TAM). To understand the technology adoption and acceptance, this model integrates the two models, which are Theory of Reasoned Action (TRA) and Technology Acceptance Model (TAM) to predict human behaviours and individual's acceptance of technology and system. This model is proved to be a better model than others, as it is useful for researchers who keen to study about the factors that determine the acceptance of technology and system. UTAUT widely applied by researchers to understand about the acceptance of system and technology (Lim et al., 2019). This UTAUT model consists of seven variables: Behavioural Intention (BI), Facilitating Condition (FC), Effort Expectancy (EE), Performance Expectancy (PE), Social Influence (SI), Security (S), Trust (T) (Abdullah et al., 2020).

2.2.3 Theory of Reasoned Action (TRA)

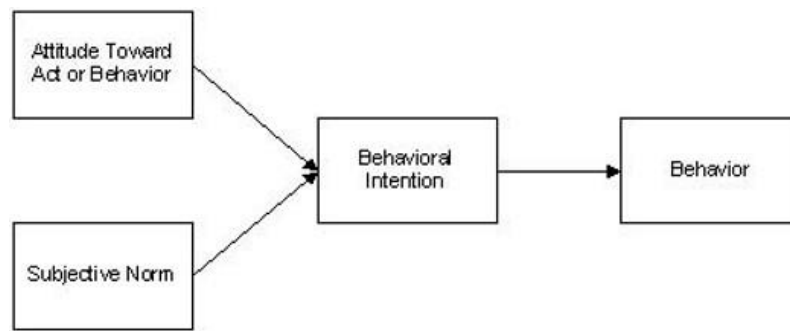


Figure 2.3 TRA Research Model (Fishbein & Ajzen, 1975)

Theory of Reasoned Action (TRA) is developed by Fishbein and Ajzen (1975). This theory is to investigate the relationship between human behaviour and attitudes. In according to TRA, behavioural intention will lead to a particular behaviour, and behavioural intention is determined by subjective norms and attitudes (Xiao, 2020). These two predictors are important to form any particular behaviour. The intention of an individual to adopt a technology or system is determined by perceived benefits and the difficulty of using the technology and system. This can be explained by TRA where an individual's beliefs will create intention and influence the attitudes to form a behaviour.

According to Sarver (1983), subjective norms refer to the perception of an individual to act under social pressure. Attitudes are belief of an individual when he or she believe that acting in certain behaviour will lead to certain outcome. Thus, if the individual believe the behaviour will lead to good outcome, then he or she will have a positive attitude on the behaviour. In contrary, if the individual assume that bad outcome will arise, then he or she will have negative attitude on the behaviour (Talukder, Quazi, Sathye, 2014).

2.3 Behavioural Intention

Behaviour intention can be defined as the willingness of an individual to adopt a new technology (Hsu & Chang, 2013). It also refers to the future behaviour of a person (Karim et al., 2020). There is literature that studied the relationship between the independent variables and dependent variables which is the behavioural intention of individuals towards services and technologies. As the user has positive perception towards using e-wallet, the intention of use will be triggered and drive the person to use e-wallet (Tan et al., 2021).

2.4 Perceived Usefulness

Perceived usefulness is defined as the extent to which an individual believe that using a particular technology system would enhance his or her job performance (David, 1989). Based on previous studies, perceived usefulness has been considered as one of the factors that influence the intention to use technologies such as digital payment, e-commerce, social media and digital banking services (Ariffin & Lim, 2020; Gupta et al., 2020; Koenig-Lewis et al., 2015).

Perceived usefulness is aim to determine the acceptance level of the consumers in adopting e-wallet. If consumers perceive that e-wallet is useful, then this can increase the intention of the consumer to adopt e-wallet. Al Nawayseh (2020) mentioned that perceived usefulness has significant impact on the consumers' intention. This is confirmed by Rahi, Khan and Alghizzawi (2020) that e-wallet providers or policy maker should focus on perceived usefulness to ensure consumers continuous intention towards using e-wallet. Tiong (2020) found out that perceived usefulness is positively related to the intention of adopting e-banking system. This is in consistent with another findings by several authors where they agreed that perceived usefulness and the continuing intention of using mobile payment system is positively related (Talwar et al., 2020). There are several studies found that perceived usefulness is a strong determinants in influencing the consumers' intention of using e-wallet (Aji & Dharmmesta, 2019; Beldad & Hegner, 2017; Rauniar et al., 2014; Venkatesh & Bala, 2008). From the research done by Punwatkarn and Verghese (2018), there is evidence to show that perceived usefulness has significant relationship on the adoption of e-wallet. Hence, hypothesis has been developed which expected that perceived usefulness has significant influence to the consumers' intention of adopting e-wallet.

H₁: Perceived usefulness has significant influence on consumers' intention to use e-wallet.

2.5 Perceived Ease of Use

Perceived ease of use refers to the extent to which the users perceived that the technology system is easy to use and user-friendly (Liao et al., 2009). Consumer's acceptance of a new technology can be perceived as a catalyst to drive higher adoption rate. Therefore, if people found e-wallet to be convenient to use, they are more likely to adopt e-wallet to make payment (Abdul-Halim et al., 2021).

Perceived ease of use is to determine the level of consumers believe that e-wallet will not be difficult to use. If consumers perceive it is easy to adopt e-wallet, then most of the consumers will probably shift to e-wallet from physical cash. If consumers assume that e-wallet is not user-friendly to adopt, then consumers probably will remain using physical money. Past studies have showed that perceived ease of use has positive effect on the attitudes towards using mobile banking services (Rahi et al., 2020; Shaikh & Karjaluoto, 2015; Munoz et al., 2016). From study by Trivedi (2016), concluded that perceived ease of use is crucial predictor to influence the adoption of e-wallet in India. This is confirmed by Pousttchi and Wiedemann (2007), where the study stated that perceived ease of use has significant effect on the intention to accept digital payment system. However, there are some researches show that the relationship between perceived ease of use and the consumers' intention to accept e-wallet is insignificant. This is proven in study by Havidz, Aima, and Wiratih (2018), where perceived ease of use is insignificant variable to influence the consumers' intention of adopting digital payment in China.

Furthermore, previous studies also examined there had a significant and positive relationship between perceived ease of use and behavioural intention to use new technologies such mobile wallet, information system, self-service technologies (Chua et al., 2020; Diatmika et al., 2016; Chen et al., 2009; Jackson et al., 2007). Based on these previous studies, strong evidence proves that perceived ease of use is one the influential factors in explaining the intention of e-wallet usage among users. Hence, hypothesis established which assuming that perceived ease of use has significant influence on consumers' intention to use e-wallet.

H₂: Perceived ease of use has significant influence on consumers' intention to use e-wallet.

2.6 Perceived Security

Perceived security is defined as the perception of a person to the extent of whether a particular technology system is safe and secure (Pantano et al., 2012). One of the main factor that cause Malaysian to be hesitate to use e-wallet is the security concern (Ismail, 2021). Thus, in this study, perceived security is deemed to be the most critical variable to be examined.

According to Ahmad et al. (2010), due to lack of experience of using new technology, the growth of technology advancement has affected users' willingness to disclose their personal financial details though the internet or smartphone application. This has emphasized that perceived security is one of the significant concern that affect people intention to use e-wallet.

From the study conducted by Soodan and Rana (2020), perceived security and the intention of adopting e-wallet is positively related. Siau and Shen (2003) also agreed that perceived security risk is the barriers to achieve high e-wallet adoption rate because consumers tend to worry about their private information will be exploit.

Cybercriminals existing in various tactics to scam e-wallet users. For instance, the user might accidentally clicked on a phishing link, which consequently installing a spyware into their phone, thereby allowing the hacker to see their password and transaction authorisation code (TAC) which is sent through text (Tan, 2022). Nielsen Payment Landscape Report also mentioned that security issue is the main barrier that deter Malaysian to adopt the usage of e-wallet, where 46% of the non-users stated that security concerns is their biggest barrier to adopt e-wallet. About 59% of the non-users believe that e-wallets can lead to credit and debit card fraud, while 38% are concerned about missing transactions through e-payment (Tan, 2019).

Nevertheless, many researchers found that the perceived security has positive impact on the intention to accept new technologies. According to Dinev et al. (2008), perceived security has a major influence on consumers' intention. Previous studies also found that perceived security will influence consumers' intention significantly (Karim et al., 2020; Humbani, 2018; Nizam et al., 2018). There are several studies found that the relationship between perceived security and intention to use e-wallet is insignificant (Chua et al., 2020; Chern et al., 2018). Aslam, Ham and Arif (2017) claimed that the perceived security has no correlation on intention to adopt e-wallet. Hence, this research has to determine whether perceived security could influence the consumers' intention towards e-wallet. Hence, this research intent to identity the relationship between perceived security and consumers' intention to use e-wallet.

H3: Perceived security has significant influence on consumers' intention to use e-wallet.

2.7 Subjective Norm

Subjective norm is defined as the perception of an individual to act under social pressure (Sarver, 1983). In the study of conformity experiment, which proposed by Solomon Asch (1958), it is believed that social pressure will have influence on a person's mind or opinion (Larsen, 1974). Such pressure is deemed to be coming from peers to influence the opinions and beliefs, where the individual is in social network (Talukder, Quazi & Djatikusumo, 2013).

Based on the Theory of Reasoned Action (TRA), subjective norm is deemed to have direct impact to the behavioural intention of an individual (Shatskikh, 2013). Aslam, Ham and Arif

(2017) also agree that subjective norm is a significant determinant to the adoption of e-wallet. Ajzen (1991) mentioned that one's behavioural intention will be influenced by peers. In addition, according to a study from Nanjing, China, it is stated that subjective norm is a strong predictor on influencing the intention of consumers in accepting e-wallet (Teng et al., 2018). Subjective norm also proved that it has significant influence and positive impact in shaping the user's satisfaction (Alalwan, 2020; Hsiao et al., 2016).

Past research conducted by Abrahamson and Rosenkopf (1997), also found that subjective norm will have large impact on potential adopters to adopt the technological innovation. Venkatesh and Davis (2000) also stated that there is a significant relationship between subjective norm and intention of an individual to use e-wallet. A study done by Yang et al., (2012) also concluded that peers such as friends, colleagues and family will have significant influence on an individual's intention to use e-wallet. Subjective norm is deemed to be an important variable to determine the consumers' intention to use since the peers of the individual is likely to influence the intention (Alalwan et al., 2017). Thus, hypothesis is established assuming that subjective norm has significant influence on consumers' intention to use e-wallet.

H4: Subjective norm has significant influence on consumers' intention to use e-wallet.

2.8 Attitude

An individual's attitude is referred to the degree to which a person feels positive or negative when performing the target behaviour (Abdul-Halim et al., 2021; Liao et al., 2009). According to study done by Wu and Chen (2017), there is a positive relationship between consumers' attitude and consumers' intention to adopt e-wallet. Hence, it is believed that individual's attitude is an important determinant in influencing individual's intention in adopting e-wallet (Foroughi et al., 2019).

Additionally, Weng et al. (2017) also indicated in previous studies that attitude have a positive correlation the intention to use mobile wallet. For consumers that exhibit positive perception towards e-wallet are enthusiastic about accepting and adopting it (Cheng et al., 2019). The behavioural intention will high as the users have highly positive attitude in adopting the new technology (Chuang et al., 2016).

Prior studies on technology adoption consider the concept of attitude, in which it reflects the favourable or unfavourable feelings of an individual about a particular behaviour and it is an

important determinant in influencing the intention of adoption (Yang et al., 2017; Apanasevic et al., 2016). Hence, hypothesis is established assuming that subjective norm has significant influence on consumers' intention to use e-wallet.

H₅: Attitude of people has significant influence on consumers' intention to use e-wallet.

2.9 Theoretical Framework

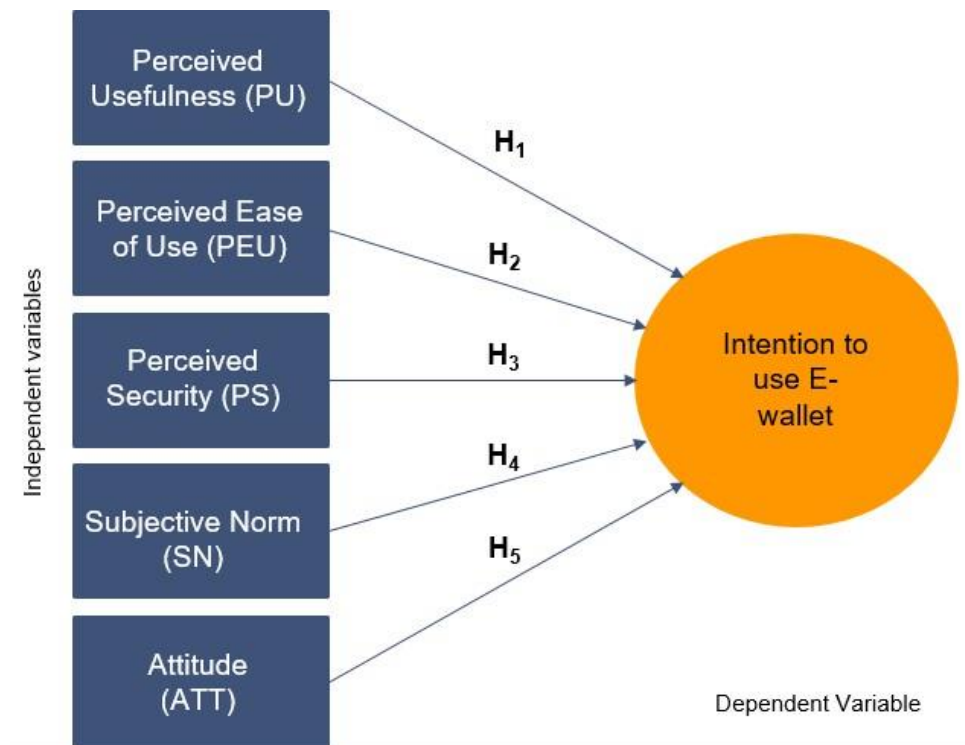


Figure 2.4 Conceptual Framework

(source: constructed by author)

H₁: Perceived usefulness has significant influence on consumers' intention to use e-wallet.

H₂: Perceived ease of use has significant influence on consumers' intention to use e-wallet.

H₃: Perceived security has significant influence on consumers' intention to use e-wallet.

H₄: Subjective norm has significant influence on consumers' intention to use e-wallet.

H₅: Attitude of people has significant influence on consumers' intention to use e-wallet.

2.10 Summary

The second chapter has discussed regarding the literature reviews, development of hypotheses and theoretical framework about the consumers' intention of using e-wallet. Under the literature review part, total three theories are discussed, 5 independent variables and 1 dependent variable have been identified and explained comprehensively.

CHAPTER 3

3.0 Research Methodology

3.1 Introduction of the Chapter

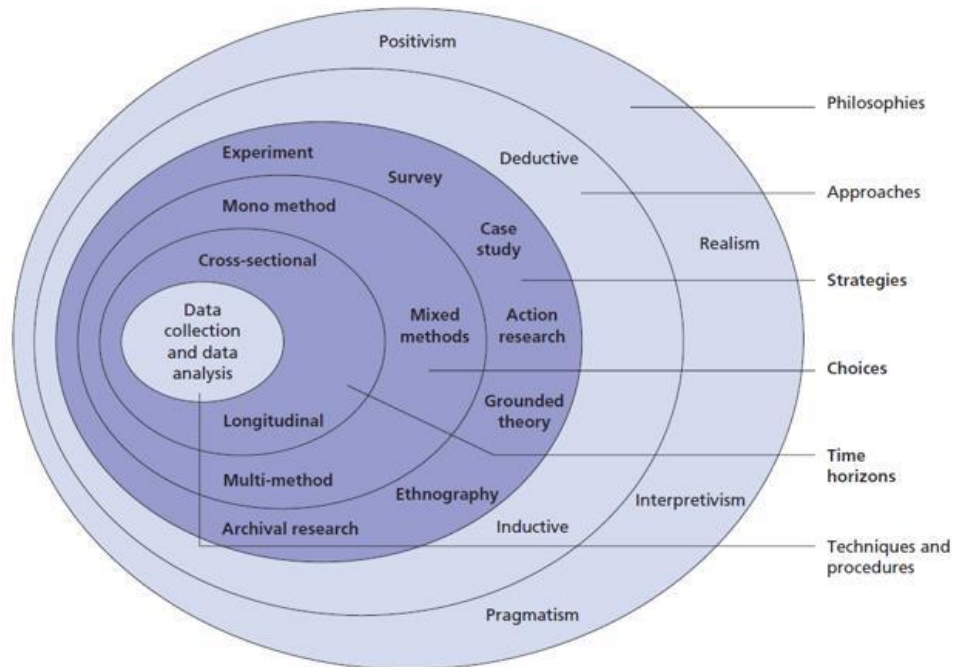


Figure 3.1 Research Onion (Saunders et al., 2009)

Research methodology refers to a general research strategy which outline how research should be undertaken. The process includes the philosophical assumptions that shape the understanding of the research question and support the choice of the research methods. Saunders et al. (2012) has proposed a theoretical concept, “Research Onion” (figure 3.1), which is to provide a simplified description about the research methodology construction in order to formulate effective methodology. Research methodology starts with the main philosophy, followed by the approach of theory development, methods, strategies and time horizons, which in the end lead to research design, the main procedures of data collection and analysis (Melnikovas, 2018).

In this chapter, methods used to conduct this research and procedures will be explained. It includes research design, research philosophy, research approach, research strategy, methodological choice, time horizon, data collection methods, sampling design, operationalization and construct measurement, data processing and analysis, pilot study.

3.2 Research Design

The research design is known as the general plan that outline the collection, measurement and analysis of the data to answer the research question for the study (Akhtar, 2016; Saunders et al., 2009). In other words, it is referred to a systematic plan that illustrate the procedures in collecting and analysing the data to answer the question for this research (Maxwell, 2012). It considered the objectives, time span and the targeted respondents for the study. Additionally, it is adopted to collect relevant data and plan the method to analyse the data. Research design is essential because it facilitates a variety of research procedures to conduct the research smoothly. It is necessary in conducting a research to gather maximum information or data with minimum expenditure of time, money and effort (Akhtar, 2016).

3.3 Research Philosophy

Research philosophy refers to the assumptions in which the research strategy and methods are based on. Business and management researchers concern about the philosophical commitments when choosing the research strategy because research philosophy not only have significant impact to what researchers did, but also what the researchers are investigating about. There are four types of research which are pragmatism, positivism, realism and interpretivism, where different research philosophies have different data collection methods (Saunders et al., 2009).

Positivism research philosophy will be used in this research. It is highly structured and the researcher need to collect a large sample to measure by using the quantitative method. By adopting this positivism philosophical stance, researchers will use the existing theory to develop hypotheses. These hypotheses will be tested and confirmed through using a research strategy to collect data (Remenyi et al., 1998).

3.4 Research Approach

There are two types of research approaches, which refers to deductive approach and inductive approach. Different type research approaches are related to different research philosophies, where deductive approach is associated to positivism research, whilst inductive approach is owed to interpretivism research. Under deductive approach, it allow researcher to test the hypotheses, where researchers are required to deduce the hypotheses from the study so as to interpret the causal relationship between the constructs. One of the important characteristics of deductive approach is that the constructs needed to be operationalized and measure

quantitatively. In order to generalise the social behaviour of human, sufficient number of samples sizes have to be selected for the study (Saunders et al., 2009).

In this research, the deductive approach of research is adopted. Based on the dependent variable, which is the intention to use e-wallet, and the independent variables, which are perceived usefulness, perceived ease of use, perceived security, subjective norm and attitude, instrument has designed to investigate the relationship between these variables.

3.5 Research Strategy

There are different types of research strategies available, for examples, survey, experiment, cash study, action research, grounded theory, ethnography and archival research. It is crucial that researchers apply the suitable strategy that will can meet objective of the research and answer the research questions. Thus, researchers should choose the research strategy based on research objective, research questions, the extent of time for the project and other resources available to undertake the research (Saunders et al., 2009).

In this study, online survey will be conducted, in which questionnaires will be constructed through Google Form to collect the data from the target respondents. Electronic survey is used in this study because it is considered as the most convenient and efficient way to collect necessary information (Sekaran & Bougie, 2016). The online survey will be distributed to a total of 245 respondents through social media platforms, for examples, WhatsApp and LinkedIn, and email to the participants. The structure of the questionnaire is made up of two sections, which they are section A and B. Section A is to collect the demographic information of the participants, whilst section B is to measure all the six variables. All items were measured through a five-point Likert scale ranging from one to five (1=Strong Disagree, 5=Strongly Agree).

3.6 Methodological Choice

The research choice can be classified as two categories, in which they are mono method and multiple methods. Mono method refers to using single data collection technique and analysis procedures to carry out the research whilst, multiple methods imply that more than one data collection technique and analysis procedures can be used to answer the research question. Under mono method, researchers can either combine single quantitative data collection

technique, for example, questionnaires, with the quantitative data analysis procedures; or single qualitative data, for instance, interviews, with the qualitative data analysis procedures. In contrast, multiple methods imply that more than one data collection techniques are used with the associated analysis procedures, in which it further break down into two methods, multi method and mixed method (Saunders et al., 2009).

In this study, mono method is used as the research choice for this study, where the only data collection method is using questionnaire to collect the data for the research. Questionnaire is categorized as the quantitative type of data collection technique, and the corresponding data analysis procedure for this study will be discussed.

3.7 Time Horizon

In terms of time horizon, the studies can be categorised as two types, in which they are longitudinal studies and cross-sectional studies. Literally, longitudinal study is referred to study that required longer period to study the change and development. Under the longitudinal studies, researchers can benefit from gaining powerful data, which will provide valuable insights. However, due to time constrained, cross-sectional type of study is widely used to conduct for research projects undertaken for academic projects.

Thus, in this research, cross-sectional study is conducted since short period of time is available to carry out the research (Saunders et al., 2009). Under cross-sectional studies, survey is the data collection technique that typically adopted (Easterby-Smith et al., 2008; Robson, 2002).

3.8 Data Collection Methods

Data collection process is an important stage in researching, and it is known as the process to gather information and measure the variables. The aim of data collection is to obtain quality evidences and transform to data analysis to establish credible and convincing answer to the question for the study. To maintain data integrity, accurate data shall be obtained (Kabir, 2016).

Quantitative method of data collection rely on random sampling and structured data collection instruments. This data collection method produce results that are easy to summarise, make comparison and generalise (Kabir, 2016).

3.8.1 Primary Data

Primary data refers to data that has not been published before and is collected as first-hand information. Primary data is more authentic, reliable, objective type of information and it has a greater validity than secondary data. Primary data may be difficult to obtain due to the scarcity of population and lack of cooperation. Sources of primary data includes questionnaire, survey and interview (Kabir, 2016).

For this research, survey is used to collect the primary data, in which survey is a widely used data collection method in management, marketing, social sciences, and psychology. The most commonly used method in survey is questionnaire. It consists of a list of questions in either open-ended or close-ended form for respondents to answer (Kabir, 2016).

Due to the fact that social distancing rule imposed as a result of COVID-19, online survey tool, Google Form will be adopted to collect data. Besides that, online questionnaire not only can reduce time and costs but also environmental friendly since no paper will be needed. In this study, the time period for collecting the data is from 25th January 2022 to 10th March 2022.

3.8.2 Secondary Data

Secondary data refers to data that has already published, in which they can be in different forms, such as journal articles, books, newspapers, biographies and databases. Although secondary data may not be as high validity as primary data, however, when it is difficult to collect primary data, or when the respondents is reluctant to provide information and their thoughts in the questionnaire (primary data), secondary data can be suffice to carry out the research. One of the major benefits of secondary data is for researchers who re-use the data do not need to do much background work on the secondary data obtained because the validity and reliability of the data have already been established and examined by previous researchers. For examples, literature review, statistics and case studies. Secondary data is also helpful in research design where it provides a base line to which the primary data obtained can be compared to. Hence, research activity should begin by conducting the literature review to review the secondary data (Kabir, 2016). In this study, a comprehensive review of literature will be done to compare the current result of primary data collected.

3.9 Sampling Design Process

Sampling is known as the procedure to choose a segment from the targeted population to represent the whole population (Malhotra & Peterson, 2006). Due to financial and time constraints, it is impracticable to collect data from the entire population. Thus, a sample has to be selected for conducting the research. When the project has a tight deadline, sampling can help researchers to save time and make the data collection process more manageable (Saunders et al., 2009).

Sampling design techniques can be separated into two types, which they are probability sampling (or known as representative sampling), and non-probability sampling (judgemental sampling). Probability sampling refers to each instance has equal chance to be selected. Whereas, for non-probability sampling, it is referred to the chance of whether each instance will be selected is uncertain (Saunders et al., 2009). Hence, for this study, non-probability sampling will be adopted.

3.9.1 Sampling Technique and Type

Non-probability sampling is deemed to be the most practical method in exploratory stage of research. There are a variety of non-probability sampling techniques available for researchers to choose based on the subjective judgement. For instances, convenience sampling, quota sampling, purposive sampling, self-selection sampling and snowball sampling (Saunders et al., 2009).

In this study, convenience sampling is being adopted as the sampling technique. It is also known as haphazard sampling, in which the samples that are easier to obtain are selected haphazardly to form the sample (Saunders et al., 2009). It is useful for researchers, when they are facing constraints in term of time and resources (Etikan, Musa, & Alkassim, 2016). Under convenience sampling, respondents that have been chosen have actually met the practical situations such as greater accessibility and availability. In other words, the group of respondents with easier accessibility will be selected for this study. In this study, convenience sampling will be used to select the respondents from the entire Malaysian population for this study.

3.9.2 Sampling Size

According to Sekaran and Bougie (2010), sample size is known as the subset of the population in a study. It is referred to the number of individual samples that are selected for conducting the survey. The sample size for this study is 245 respondents. Draft questionnaire will be send to 50 respondents to conduct the pilot test in order to demonstrate that all constructs can be properly utilize.

3.9.3 Target Population

The purpose of this study is to examine the intention of Malaysian consumers in adopting e-wallet. Thus, the target population for this study will be respondents who is Malaysian, and above 18 years old, regardless of gender and race who have the intention or experience in using e-wallet in Malaysia. For the mature group of respondents (from 26 years old and above), working professional, housewife, and retiree will be the targeted as most of these respondents have at least one mobile device. For young respondents (from 18 years old to 25 years old), the respondents can be students (from high school to university). The reason to targeting the young respondents is because they are technology savvy and are fast in adopting new technologies.

3.10 Operationalization and Measurement of Constructs

Constructs	Operationalization	Scale	References
Perceived usefulness	6 items to measure continued intention as follows: 1. E-wallet is a useful mode of payment. 2. Using E-wallet would enhance my effectiveness in making payment. 3. Using E-wallet can save my time. 4. Using E-wallet allow me to pay quickly. 5. Using E-wallet would improve my shopping experience as a consumers. 6. Using E-wallet can improve my performance in managing personal payments.	Five-point Likert Scale (ranging from 1=strongly disagree and 5=strongly agree)	Adapted from Daragmeh et al. (2021)
Perceived ease of use	5 items to measure continued intention as follows: 1. E-wallet is easy to use. 2. Understand the steps required to use E-wallet is easy 3. I rarely get confused when I use E-wallet. 4. It is easy to perform the steps required to use E-wallet. 5. I rarely make errors when using the E-wallet.	Five-point Likert Scale (ranging from 1=strongly disagree and 5=strongly agree)	Adapted from Daragmeh et al. (2021)

Perceived security	6 items measure continued intention as follows: 1. I feel secure with using E-wallet to make payment. 2. I feel comfortable with having my E-wallet integrated into my mobile phone 3. The risk of an authorized third party overseeing the payment is low 4. The risk of abuse of billing information is low 5. The risk of abuse of usage (i.e. payment amount) is low 6. I believe that overall riskiness of using E-wallet is low.	Five-point Likert Scale (ranging from 1=strongly disagree and 5=strongly agree)	Adapted from Rahman et al. (2021)
Subjective norm	4 items measure continued intention as follows: 1. People who are important to me believe that using E-wallet is beneficial. 2. People who are important to me would consider using E-wallet to be a good idea. 3. People who are important to me would recommend using E-wallet. 4. It is expected that people that are similar to me would use E-wallet.	Five-point Likert Scale (ranging from 1=strongly disagree and 5=strongly agree)	Adapted from Flowers et al. (2017)

Attitude	4 items measure continued intention as follows: 1. I like the idea of using E-wallet for transactions. 2. It is a wise idea to use E-wallet for transactions. 3. Using E-wallet for payment is beneficial. 4. Using E-wallet is time-saving.	Five-point Likert Scale (ranging from 1=strongly disagree and 5=strongly agree)	Adapted from Abdul-Halim et al. (2021)
Intention to use E-wallet	4 items to measure continued intention as follows: 1. I intend to use E-wallet in my daily life. 2. If I could, I would use E-wallet as much as possible in the future. 3. I would continue to use E-wallet in the future. 4. I intend to recommend others to use E-wallet.	Five-point Likert Scale (ranging from 1=strongly disagree and 5=strongly agree)	Adapted from Alraimi et al. (2015)

Table 3.1: Operationalization and Measurement of Constructs

(source: constructed by author)

Table 3.1 shows that the list of construct measurements, with total 29 items. For the measurement scale, Five-point Likert scale will be adopted to measure the each items in the questionnaire. It is ranged from strongly disagree=1, disagree=2, neutral=3, agree=4, strongly agree=5.

3.11 Data Analysis Method

After collecting the primary data from the questionnaires, the data will be analysed through SMART PLS 3 software. To analyse the outcome of the model, Partial Least Squares Structural

Equations Modelling (PLS-SEM) will be used as the sampling technique. The reason SMART PLS 3 software is used is because of its strength where reliability and validity results can be constructed with small sample sizes. Other than this, SMART PLS 3 can modify the abnormal data with the central limit theorem to maximize the r square and minimize the errors.

3.12 Reliability

One of the most important elements to test the quality of the data is reliability, in which it is concerned with whether the data collection techniques and analysis procedures can generate consistent findings (Easterby-Smith et al., 2008). There are several methods to test the reliability, including internal consistency, decision consistency, test-retest reliability, parallel forms reliability, inter-rater reliability, odd-even reliability, split-half methods and alternative forms methods (Fisher, 1992).

3.12.1 Internal Consistency Reliability

Internal consistency is known as a measurement based on the degree of bivariate correlations between different items on the same test (Cronbach, 1951). It aims to measure whether the items under the same construct lead to consistent scores on a test (Tang, Cui, & Babenko, 2014). Cronbach's alpha is the conventional criterion for internal consistency, in which it provides reliability estimation based on correlations within the indicator variables. However, due to Cronbach's alpha may underestimate the internal consistency reliability because it is sensitive to the number of items on the scale. To overcome the weakness of Cronbach's alpha, composite reliability is another criterion that is more appropriate to apply to measure the internal consistency reliability (Saunders et al., 2009).

For Cronbach's alpha, a value of 0.7 or higher are preferable (Wong, 2013). Whilst for composite reliability, value that is higher than 0.7 and lower than 0.95 is acceptable. Generally, if the result obtained shows higher value, indicates that level of reliability is high (Saunders et al., 2009).

3.13 Validity

Validity refers to whether or not the findings are truly about what they claim to be about. It is to investigate there is a causal relationship between the variables (Saunders et al., 2009). It can

also be defined as the degree to which the variables are appropriately measured in the quantitative research (Heale & Alison, 2015).

3.13.1 Convergent Validity

Convergent validity refers to the degree to which a measure correlates with the alternative measures within the same construct positively (Hair et. al., 2017). It is also known as a measure of construct converges so as to examine the variance of the construct items. To evaluate the convergent validity of the variables, researchers have to consider the outer loadings of the framework and average variance extracted (AVE).

The outer loading should have a value that is larger than 0.70 to satisfy the convergent validity. Besides that, the value should greater than 0.50 to confirm sufficient convergent validity (Hair et al., 2018).

3.13.2 Discriminant Validity

Discriminant validity aims to measure the extent of whether a particular variable is empirically distinct from other variables (Hair et al., 2017). There are several methods that were applied to assess the discriminant validity of the constructs. First, the widely used method to evaluate discriminant validity is using Fornell-Larcker criterion. Furthermore, cross loadings of the constructs are also adopted to examine the discriminant validity. Additionally, Heterotrait-Monotrait (HTMT) also adopted to determine the discriminant validity of the constructs.

3.14 Pilot Study

Pilot study is a crucial part in research design (Malmqvist et al., 2019). One of the main purpose to conduct a pilot study is to improve the quality and efficiency of the research (In, 2017). Researchers able to evaluate the reliability and validity of the variables by conducting a pilot study. Pilot testing the data can also help researchers to ensure the data collected enable the research questions to be answered (Saunders et al., 2009). Generally, the sample size for pilot testing is 30 samples or greater.

Variables	Average Variance Extracted (AVE) (>0.5)	Composite Reliability (CR) (0.7<CR<0.95)	Cronbach's Alpha (CA) (0.7<CA<0.95)
Perceived Usefulness (PU)	0.670	0.924	0.901
Perceived Ease of Use (PEU)	0.630	0.894	0.851
Perceived Security (PS)	0.792	0.950	0.934
Subjective Norm (SN)	0.802	0.942	0.917
Attitude (ATT)	0.787	0.936	0.909
Intention of Use (ITU)	0.765	0.928	0.896

Table 3.2: Result of Average Variance Extracted (AVE), Composite Reliability, Cronbach's Alpha for 50 samples. (source: constructed by author)

In this study, 50 samples are used to run the pilot test. As shown in table 3.2, the average variance extracted value of all variables are greater than 0.5. Besides that, each of the variables also generated a Cronbach's alpha value that is ranged between 0.7 and 0.95. The composite reliability of all the variables have presented a value that is higher than 0.7 but below 0.95. Hence, as the results of the pilot test shows that all the variables have satisfied the criteria of average variance extracted, composite reliability and Cronbach's alpha, indicates that the variables are reliable measurement for the study.

3.15 Research Ethics

Ethics is one of the concerns during research process. Research ethics refers to how the researcher formulate the research topic, establish the research design, gain access to data sources, collect data, analyse the results and present the research findings in an ethical and responsible manner. The conduct of a researcher is guided by code of ethics which outline a set of ethical guidelines to govern the behaviour of the researchers and prevent misconduct (Saunders et al., 2009). According to American Psychological Association (APA), plagiarism is deemed to be one of the ethical issues. As an ethical researcher, one should not credit

another's research work or data as their own and all the data and resources used in the study should be cited appropriately.

Besides that, researchers should obtain the informed consent from the participants during data collection stage. The informed consent should include the purpose of the research and the expected duration of the questionnaire. Researchers should also mentioned that the participants have the right to withdraw from the questionnaire since the participation is entirely voluntary. All the participants' information should be confidentially. Contact information should be stated to provide the participants with the opportunity to make enquiries if they have any (American Psychological Association, 2017). The research ethics form for this study is attached in the appendices. Please refer to appendix c.

3.16 Limitation of the Methodology

For this study, there are some limitations in the research design and methodology. First, only a single data collection method that is employed for this study, which it is the questionnaire. According to Saunders, Lewis and Thornhill (2009), using different data collection technique will lead to different result. Therefore, different techniques and procedures shall be adopted for data collection to eliminate the 'method effect'. Many research books also suggested to collect the qualitative data continuously, such as to conduct additional interviews with the respondents. Nevertheless, due to the ongoing pandemic situation, it would be difficult to carry out interview with the respondents.

Smith (1981) also stated that quantitative and qualitative data collection methods have its own strengths and weaknesses. The drawback of employing online questionnaire to collect data, researcher may face the issue of losing control in overseeing the respondents, whether they have answered the questions honestly and properly. If the respondent did not answered the questions properly, this may distort the reliability of the result for the study.

Furthermore, due to time constraints of the project, this may limit the time span to collect the data through other method such as interview, to get higher response rate and enable respondents to express their thoughts and feelings (Alshenqeeti, 2014). In this study, convenience sampling technique is adopted. One of the disadvantages of using convenience sampling method is that it has high degree of bias in selection of population (Galloway, 2005). Even though convenience sampling technique is widely used by many researchers because it is easier way to obtain the sample, it will introduce bias to the sample (Saunders et al., 2009).

3.17 Summary

In this entire chapter, the methodology of the research is explained comprehensively. Online questionnaire is used as the method of collecting data in this study. The target population for this study is people in Malaysia who is 18 years old and above.

CHAPTER 4

4.0 Data Analysis

4.1 Introduction of the Chapter

In this chapter, the data collected from the questionnaire will be analysed and interpreted. By using SMART PLS 3 Software to compute the data, descriptive analysis of the final result will be presented in this chapter. This chapter will also analyse the demographic profile of the respondents.

4.2 Descriptive Analysis

Descriptive analysis refers to the use of descriptive statistics to summarise the data collected in an organized manner (Kaur et al., 2018). Descriptive statistics also enable researchers to compare and interpret the variables numerically so as to make the data more understandable by the audiences (Saunders et al., 2009).

In this study, pie charts and diagrams are used to explain the demographic data of the respondents by showing the frequency and percentage data of the 245 respondents who have participated in the questionnaire.

4.2.1 Demographic Statistics of Respondents

In this section, demographic information of the respondents will be presented. There are total 245 respondents take part in this research. Participants' consents have been sought before respondents filling out the questionnaire, which implies that 245 respondents agree to provide their information and believed that the information will be kept confidentially.

4.2.1.1 Gender

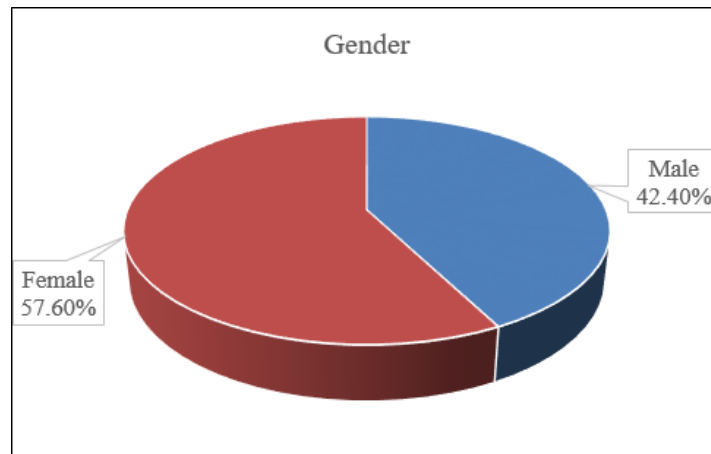


Figure 4.1 Gender Distribution

(source: constructed by author)

According to figure 4.1, the pie chart shows that there are total 245 respondents participated in this questionnaire. Among the respondents, 104 of the respondents are male, which is 42.4% of the population, whilst 141 of the respondents are female, which is 57.6% of the population.

4.2.1.2 Age

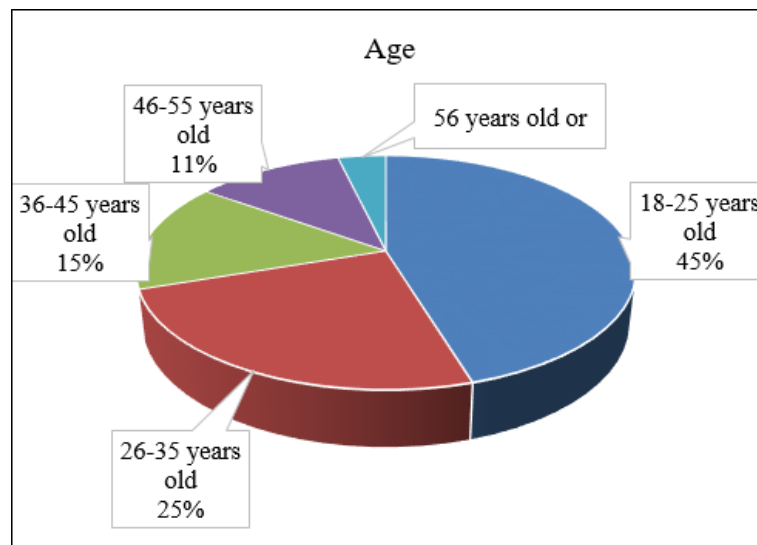


Figure 4.2: Age Group

(source: constructed by author)

Among the respondents of the questionnaire, there are 111 respondents whose age ranged between 18 to 25 years old. As illustrated in figure 4.2, the age group of 18 to 25 years old is the largest with a portion of 45%. On the other hand, the age group of 56 years old or above has the least number of respondents with 9 respondents, comprised to only 4%. There are 60 respondents whose age ranged between 26 to 35 years old, amounted to 24.5% of the population. Followed by age group of 36 to 45 years old and 46 to 55 years old, which comprised of 37 respondents (15%) and 28 respondents (11%) respectively.

4.2.1.3 Ethnicity

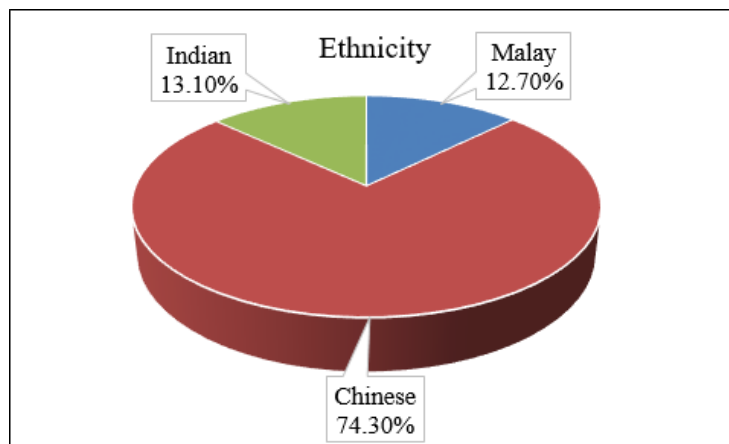


Figure 4.3: Ethnic Group

(source: constructed by author)

Figure 4.3 presented the ethnicity of the respondents, where the highest number of ethnic group is Chinese, consists of 182 respondents, amounted to 74.3%. The second largest ethnic group in this questionnaire is Indian comprised of 32 respondents with 13.1% of the population. Number of Malay respondents are slightly lower than Indian respondents, with 31 respondents, accounted for 12.7% of the population.

4.2.1.4 Education Level

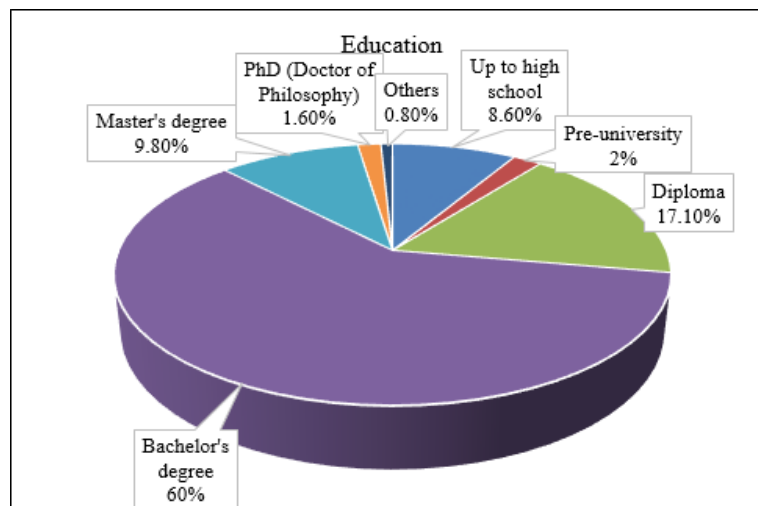


Figure 4.4: Education Level

(source: constructed by author)

According to figure 4.4, most of the respondents have Bachelor's degree, with 147 respondents, which comprised of 60% of the population. Next, there are 42 respondents who studied Diploma for their tertiary education, comprised of 17.1% of the population. 24 respondents (9.8%) have Master's degree, whilst 4 respondents (1.6%) who have obtained a doctorate (PhD). In this questionnaire, there are 21 respondents (8.6%) whose education level is up to high school, and 5 respondents still pursuing their pre-university studies, which amounted to 2% of the population. Lastly, 2 respondents fall under other education level, as one of the respondents is an ACCA affiliate and the education level of another respondent is until SPM.

4.1.2.5 Occupation

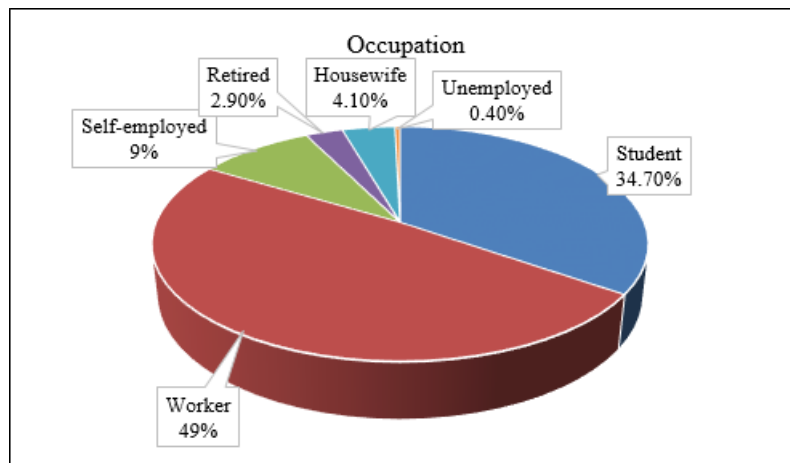


Figure 4.5: Occupation Group

(source: constructed by author)

As graphed in figure 4.5, the occupation of the largest group of respondents is workers, in which there are total 120 respondents, accounted for 49% of the population. Furthermore, 85 respondents who participated in the questionnaire are students, which they make up of the second largest occupation group with 34.7% of the population. Figure 4.5 also stated that 9% of the respondents are self-employed and 4.1% are housewife, comprised of 22 respondents and 10 respondents respectively. Moreover, there are 7 retired respondents (2.9%) and only one respondent (0.4%) is currently unemployed.

4.1.2.6 Monthly Income/Allowance

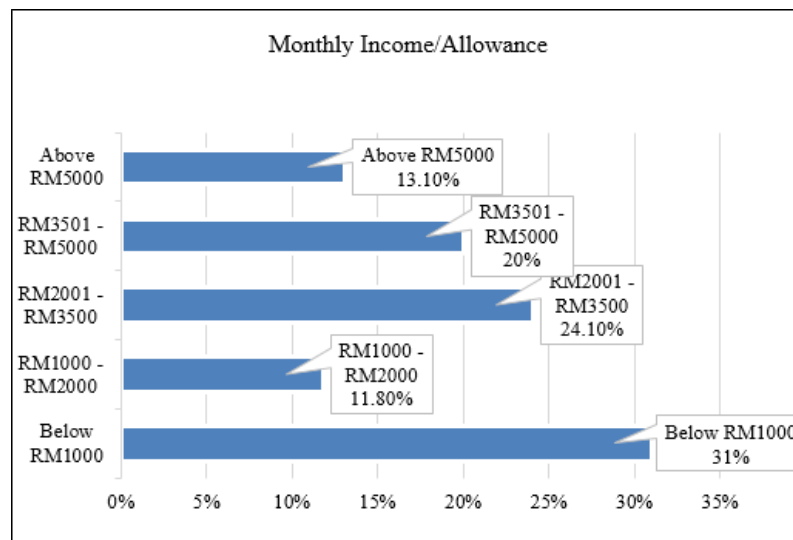


Figure 4.6: Monthly Income/Allowance Distribution

(source: constructed by author)

Figure 4.6 illustrated the monthly income or allowance distribution of the respondents. Among the 245 respondents, most of the respondents earn an income or received allowance that is below RM 1000 on monthly basis, with 76 respondents which amounted to 31% of the population. The respondents who have a monthly income or allowance ranged between RM 2001 to RM 3500 accounted for the second largest group in figure 4.6, with 59 respondents, 24.1% of the population. For the group that earn a monthly income or allowance of RM 3501 to RM 5000, comprised of 49 respondents (20%). Followed by monthly income or allowance group of RM 1000 to RM 2000 and above RM 5000, have 29 respondents (11.8%) and 32 respondents (13.1%) respectively.

4.1.2.7 Awareness of E-Wallet

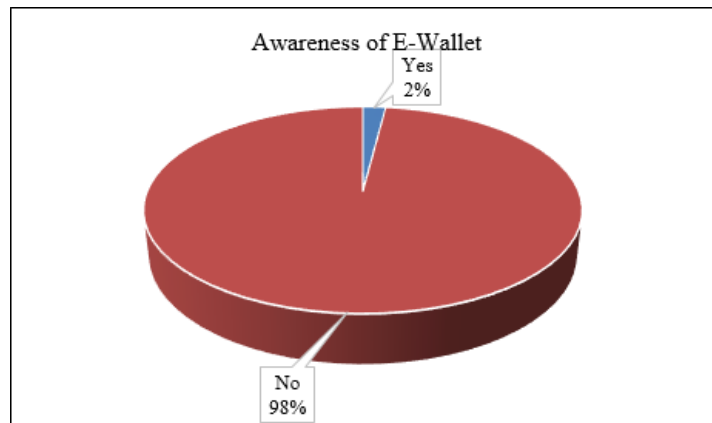


Figure 4.7: Awareness of E-Wallet

(source: constructed by author)

Among 245 respondents in the questionnaire, 98% of the respondents, which there are 240 respondents selected 'YES', indicates that most of the respondents are aware that E-Wallet exists in the society. On the other hand, only 5 respondents who are not aware of the existence of E-Wallet in the market, which comprised of 2% of the population.

4.1.2.8 E-Wallet Usage

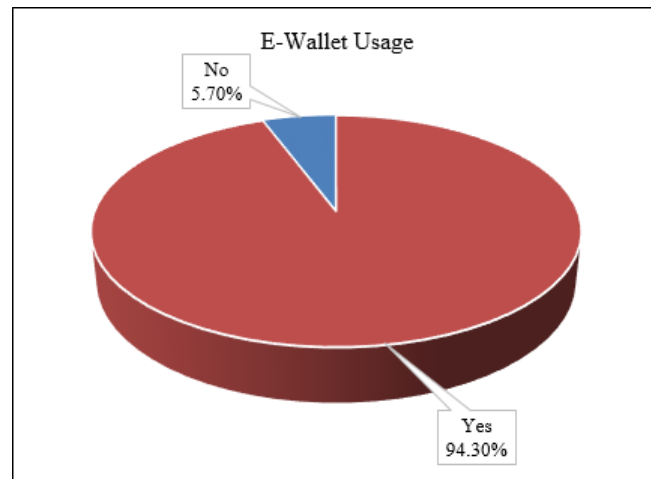


Figure 4.8: E-Wallet Usage

(source: constructed by author)

Figure 4.8 shows that there are 94.3%, which is 231 respondents, who are the E-Wallet users. On the contrary, 5.7% of the respondents, in which 14 respondents have answered that they are not E-Wallet users, and have no experience in using this type of payment method.

4.1.2.9 Frequency of E-Wallet Usage

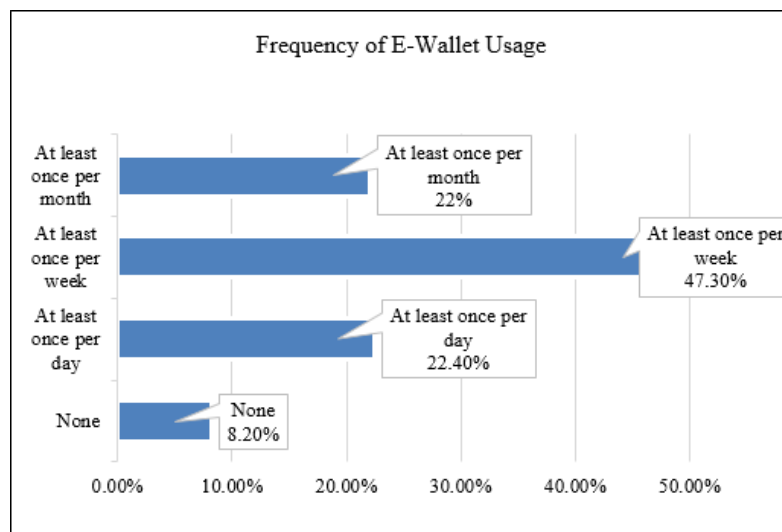


Figure 4.9: Frequency of E-Wallet Usage

(source: constructed by author)

As illustrated in figure 4.9, most of the respondents use the E-Wallet at least once per week, in which there are 116 respondents fall under this category and accounted for 47.3% of the population. The number of respondents that use E-Wallet at least once per day is 55 respondents (22.4%), which is slightly higher than the number of respondents that use E-Wallet at least once per month, with 54 respondents (22%). On the other hand, there are 20 respondents who have not use E-Wallet before, which comprised of 8.2% of the population.

4.1.2.10 Types of E-Wallet Used

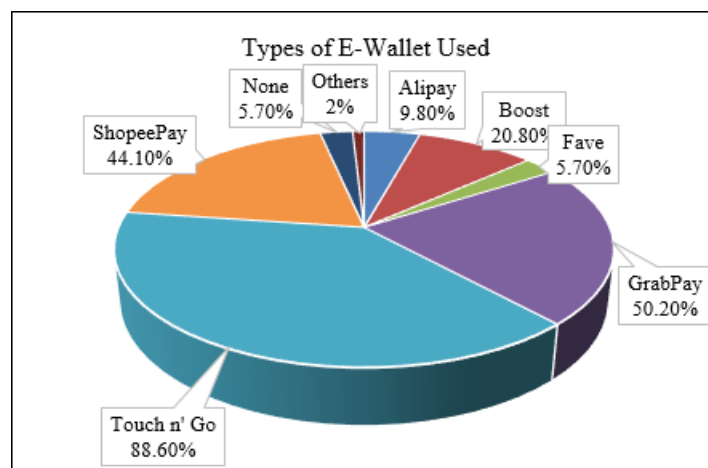


Figure 4.10: Types of E-Wallet Used

(source: constructed by author)

According to figure 4.10, the type of E-Wallet that most of the respondents use is Touch n' Go e-wallet, in which there are in total 217 respondents, amounted to 88.6% of the population. Next, the second most popular E-Wallet among the respondents is GrabPay, where 123 respondents adopted GrabPay, make up of 50.2% of the population. Followed by the third most popular E-Wallet, ShopeePay. There are 108 respondents (44.1%) use ShopeePay as their E-Wallet. As illustrated in the pie chart, 20.8% of the respondents (51 respondents) use Boost e-wallet, whilst 5.7% of the respondents (14 respondents) adopted Fave e-wallet. Lastly, there are 24 respondents adopted Alipay with 9.8% of the population. There are 5 respondents (2%) use different e-wallet other than those available in the options, for examples, Apply Pay and Samsung Pay.

4.2 Measurement Model

PLS-SEM has an aim to maximizing the variance of the dependent variable to evaluate the predictive capabilities of the model. The fundamental measurement model metrics for PLS-SEM are reliability, convergent validity, discriminant validity (Hair et al., 2017). In this section, PLS-SEM model assessment will first focusing on the measurement model to evaluate the reliability and validity of the construct measures.

4.2.1 Internal Consistency Reliability

4.2.1.1 Cronbach's Alpha, Composite Reliability, Average Variance Extracted (AVE)

The traditional criterion to examine the internal consistency reliability and validity of the study is Cronbach's Alpha (Urbach & Ahlemann, 2010). Cronbach's Alpha is aimed to examine all the variables, whether they are higher than the required threshold of 0.7 to be accepted. In this research, 245 samples have been collected to conduct the data analysis process to test whether the framework is reliable.

Variables	Cronbach's Alpha (0.7<CA<0.95)	Composite Reliability (0.7<CR<0.95)	Average Variance Extracted (AVE) (≥0.5)
Perceived Usefulness (PU)	0.905	0.927	0.680
Perceived Ease of Use (PEU)	0.872	0.906	0.661
Perceived Security (PS)	0.916	0.936	0.747
Subjective Norm (SN)	0.913	0.939	0.794
Attitude (ATT)	0.918	0.943	0.804
Intention of Use (ITU)	0.929	0.950	0.825

Table 4.1 Results of Constructs Reliability and Validity

(source: constructed by author)

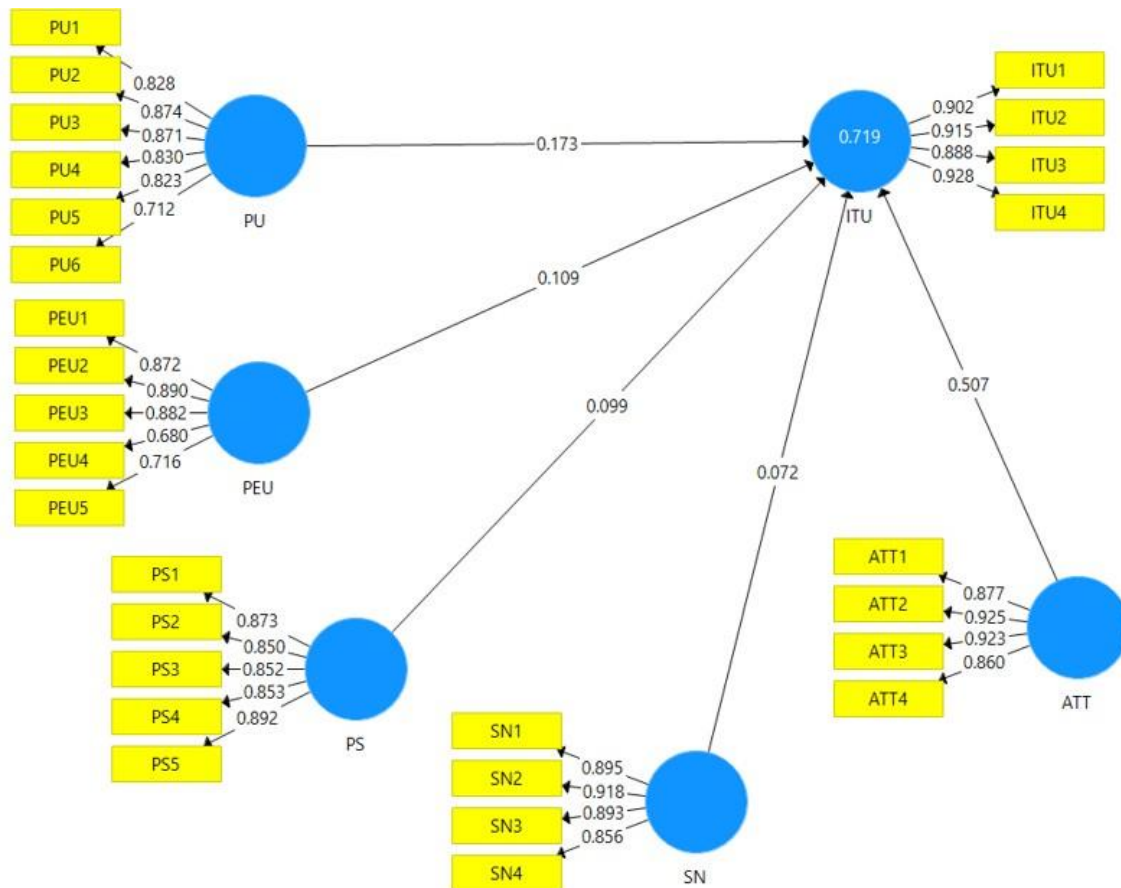
From the result shown in table 4.1, all the variables are greater than 0.7 which indicates that the value of internal constructs reliabilities is acceptable. Intention of use has the highest value of 0.929, followed by attitude with the value of 0.918. Next, perceived security is ranked as third with the value of 0.916. Subjective norms, perceived usefulness, and perceived ease of use comprised into the value of 0.913, 0.905, and 0.872 respectively.

Furthermore, the function of composite reliability is similar with that of Cronbach's Alpha. However, the result generated by Composite Reliability is more reliable than the result of Cronbach's Alpha because it take into account different outer loadings. The composite reliability varies between 0 and 1, with higher values indicating higher levels of reliability. The satisfactory range of value is between 0.7 and 0.95, variable that has the value above 0.95 will not be a valid measure.

As shown in table 4.1, all the variables have a good value of composite reliability which range from 0.906 (perceived ease of use) to 0.95 (intention of use). Thus, the result confirm that the variables in this study are highly reliable and consistent in explaining the variances constituted among each variables.

Average of variances extracted (AVE) for each construct is remained critical to the aspect of construct reliability. Fornell and Larcker (1981) mentioned that AVE should be greater than 0.5. Figure that below than 0.5 means that a measurement error was occur in the variance.

According to table 4.1, there are none of the variables that are below 50% in the AVE value. The lowest AVE value is generated by perceived ease of use with a percentage of 66.1 and the highest AVE is showed by intention of use with a percentage of 82.5. Hence, it can be concluded that the variables in the model are reliable.



(Notes: ITU=Intention to Use, PU= Perceived Usefulness, PEU=Perceived Ease of Use, PS=Perceived Security, SN=Subjective Norms, ATT=Attitude)

Figure 4.11 PLS-SEM Model

(source: constructed by author)

4.2.2 Discriminant Validity

4.2.2.1 Fornell-Lacker Criterion

Discriminant validity is observed by the variance extracted value. Fornell-Lacker Criterion is the most widely used and effective method to determine those constructs in the discriminant validity (Hamid, Sami, & Sidek, 2017). Fornell-Lacker Criterion compares AVE square root with the latent variable correlations. Each of the variable's AVE should be higher than the highest correlation with any other variables (Hair et al., 2017).

	ATT	ITU	PEU	PS	PU	SN
ATT	0.897					
ITU	0.810	0.908				
PEU	0.698	0.703	0.813			
PS	0.528	0.573	0.580	0.864		
PU	0.708	0.724	0.764	0.584	0.825	
SN	0.714	0.685	0.686	0.564	0.692	0.891

(Notes: ITU=Intention to Use, PU= Perceived Usefulness, PEU=Perceived Ease of Use, PS=Perceived Security, SN=Subjective Norms, ATT=Attitude)

Table 4.2 Fornell-Lacker Criterion

(source: constructed by author)

Table 4.2 presented the value of square root in Fornell-Lacker Criterion, where attitude has a value of 0.897. The value of 0.897 is higher than value of other variables within the same column. Similar calculation and observation is also applied for other latent variables. The result as above confirmed that the discriminant validity is well developed and this structure model can be considered as fit.

4.2.2.2 Outer Loadings

One of the measure of discriminant validity is cross loadings, in which it is the first approach to assess the discriminant validity of the constructs. To determine the discriminant validity of the constructs, outer loading of the associated construct should generate a value that is greater than cross loading of other constructs (Hair et al., 2017). All the cross loadings within each of the variables were removed to evaluate the discriminant validity of the model.

	ATT	ITU	PEU	PS	PU	SN
ATT1	0.677					
ATT2	0.925					
ATT3	0.823					
ATT4	0.860					
ITU1		0.902				
ITU2		0.915				
ITU3		0.888				
ITU4		0.928				
PEU1			0.872			
PEU2			0.890			
PEU3			0.882			
PEU4			0.680			
PEU5			0.716			
PS1				0.873		
PS2				0.850		
PS3				0.852		
PS4				0.853		
PS5				0.892		
PU1					0.828	
PU2					0.874	
PU3					0.871	
PU4					0.830	
PU5					0.823	
PU6					0.712	
SN1						0.895
SN2						0.918
SN3						0.893
SN4						0.856

(Notes: ITU=Intention to Use, PU= Perceived Usefulness, PEU=Perceived Ease of Use, PS=Perceived Security, SN=Subjective Norms, ATT=Attitude)

Table 4.3 Outer Loadings

(source: constructed by author)

Table 4.3 presented cross loading of the constructs in the study. All the outer loadings value of the items varied between 0.7 and 0.928, suggested that all the items value should be greater than 0.7. From table 4.3, all the variables have a value that is greater than 0.7, except for one of the items under perceived ease of use, in which it is slightly below 0.7, with 0.68. Overall, the results showed that the discriminant validity of the indicators can be confirmed.

4.2.2.3 Heterotrait-Monotrait Ratio (HTMT) Criterion

Heterotrait-monotrait (HTMT) refers to the ratio between the correlations of the indicators across different constructs. HTMT tend to produce greater precision as compared to other methods. This criterion is proposed by Henseler et al. (2015) to address the shortcoming of adopting Fornell-Larcker criterion to measure the discriminant validity where Fornell-Larcker criterion is poor in detecting discriminant validity issues if the indicator loadings vary widely (Hair et al., 2017).

	ATT	ITU	PEU	PS	PU	SN
ATT						
ITU	0.876					
PEU	0.759	0.760				
PS	0.558	0.606	0.627			
PU	0.774	0.788	0.838	0.630		
SN	0.773	0.738	0.750	0.606	0.760	

(Notes: ITU=Intention to Use, PU= Perceived Usefulness, PEU=Perceived Ease of Use, PS=Perceived Security, SN=Subjective Norms, ATT=Attitude)

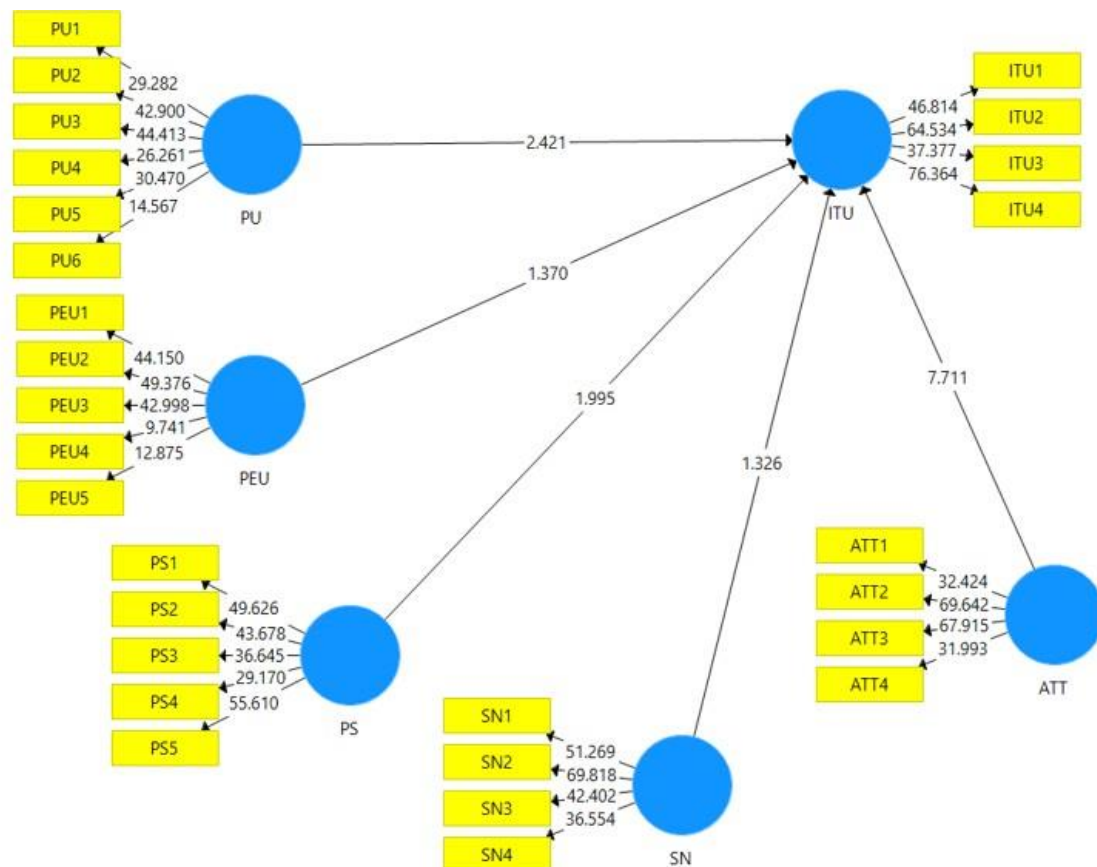
Table 4.4 Heterotrait-Monotrait Ratio (HTMT) Criterion

(source: constructed by author)

Based on prior researches and studies, it is suggested that the threshold value of HTMT to be 0.9, where HTMT value that is higher than 0.9 indicates that there the model is lack of discriminant validity. As shown in table 4.4, the HTMT value of all the constructs is below 0.9, suggested that discriminant validity of all the indicators is defined, in which the model is able to provide reliable and valid information in this study.

4.3 Structural Model Path Coefficients

After evaluating the validity and reliability of the model, PLS-SEM analysis was performed into the path coefficient. Path coefficient is one of the methods to examine the structural model. In this study, bootstrapping is used to evaluate the standard deviations, T-statistics, and P-values.



(Notes: ITU=Intention to Use, PU= Perceived Usefulness, PEU=Perceived Ease of Use, PS=Perceived Security, SN=Subjective Norms, ATT=Attitude)

Figure 4.3 Structural Model (Bootstrapping)

(source: constructed by author)

Figure 4.3 presented the structural model of this study. This model is computed based on bootstrapping method, where it has bootstrapped to 5000 samples.

Relationship	Sample Mean (M)	Standard Deviation (STDEV)	T Statistics	P Values	Result
PU -> ITU	0.178	0.071	2.421	0.016	Supported
PEU -> ITU	0.109	0.080	1.370	0.171	Unsupported
PS -> ITU	0.100	0.050	1.995	0.046	Supported
SN -> ITU	0.073	0.054	1.326	0.185	Unsupported
ATT -> ITU	0.502	0.066	7.711	0.000	Supported

(Notes: ITU=Intention to Use, PU= Perceived Usefulness, PEU=Perceived Ease of Use, PS=Perceived Security, SN=Subjective Norms, ATT=Attitude)

Table 4.5 Summary of Structural Model Path Coefficient

(source: constructed by author)

Table 4.5 presented the summary of structural model and the significant level was set at 0.05 (5%). This implies that when P-value is exceeded the significant level of 0.05, the result is unsupported. Previous study has shown that if P-value < 0.05, the model is significant (Malhora & Peterson, 2006). One of the studies also mentioned that when P-value is smaller than the significant level of 0.05, the relationship of independent variable and dependent variable is significant (Uyanık & Güler, 2013). Based on the result shown in table 4.3, the perceived ease of use (PEU) and subjective norm (SN) do not have significant relationship with the intention to use (ITU). Other independent variables, perceived usefulness (PU), perceived security (PS), and attitude (ATT) have significant relationship with the independent variable – intention to use (ITU).

4.3.1 Coefficient of Determination (R^2)

Coefficient of determination, which is also known as R^2 value, is most widely used measure to evaluate the predictive power of the structural model. According to Hair et al. (2017), the R^2 value ranges from zero to 1, in which higher value indicates higher degree of predictive accuracy.

	R Square	R Square Adjusted
Intention to Use (ITU)	0.719	0.714

Table 4.6 R Square results

(source: constructed by author)

Based on past researches under discipline of marketing, the rule of thumb for R^2 value of 0.25, 0.5, 0.75, represent weak, moderate, substantial respectively (Hair et al., 2011; Henseler et al., 2009). From table 4.6, the result showed that the 71.9% of the variance in the model can be justify, which indicates that this model has high level of predictive accuracy.

4.3.2 Effect Size (f^2)

	ATT	ITU	PEU	PS	PU	SN
ATT		0.348				
ITU						
PEU		0.014				
PS		0.021				
PU		0.035				
SN		0.007				

(Notes: ITU=Intention to Use, PU= Perceived Usefulness, PEU=Perceived Ease of Use, PS=Perceived Security, SN=Subjective Norms, ATT=Attitude)

Table 4.7 Effect Size results

(source: constructed by author)

Effect size, which is also known as f^2 value, is aim to measure magnitude of the effect of independent variables on dependent variables (Selya et al., 2012). The rule of thumb for effect size of 0.35, 0.15, 0.02, is described as large effect, medium effect, and small effect respectively.

From table 4.7, the result showed that the consumers' attitude has great effect on their intention to use e-wallet, with 0.348. Whereas, perceived ease of use and subjective norm showed no effect on consumers' intention to use e-wallet, as the effect size value showed 0.014 and 0.007 respectively.

4.4 Hypotheses Testing

4.4.1 Research Objective 1 – Perceived Usefulness

H₁: Perceived usefulness has significant influence on consumers' intention to use e-wallet

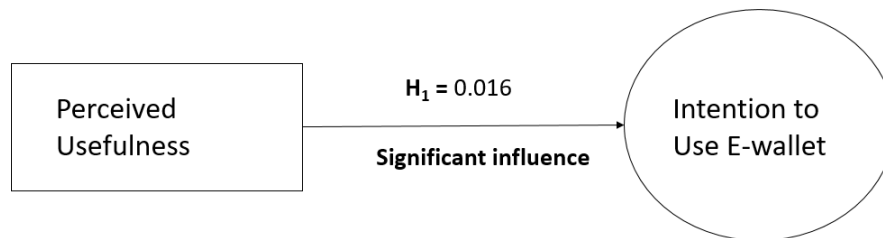


Figure 4.4 Result of Hypothesis 1

As evident in figure 4.4, the H₁ is supported by this study, where it showed that perceived usefulness is a significant variable that affect the intention to use e-wallet, because the value is below 0.05. This result is in line with the conclusion made by Ariffin et al., (2021) that perceived usefulness has positive impact on consumers' intention. According to Trivedi (2016) which had done the study in India also concluded that perceived usefulness has significant influence on consumers' intention to adopt e-wallet. Moreover, the result of this research also agree with Al-Marroof and Al-Emran (2018) study, where it is believe that there is a significant relationship between perceived usefulnesss and intention to adopt a particular technology. Kleijnen, Wetzels and De Ruyter (2004) had emphasised that perceived usefulness can justify the consumers' intention to adopt e-wallet as the payment method in their daily life. If consumers perceived that using e-wallet is beneficial, they will probably have a positive intention towards using the e-wallet to make payment (Juniwati, 2014). Hence, the result of this study has proved the Theory of Acceptance Model (TAM) proposed by Davis (1989) that perceived usefulness has an impact on the consumers' intention in adopting particular system.

4.4.2 Research Objective 2 – Perceived Ease of Use

H₂: Perceived ease of use has significant influence on consumers' intention to use e-wallet.

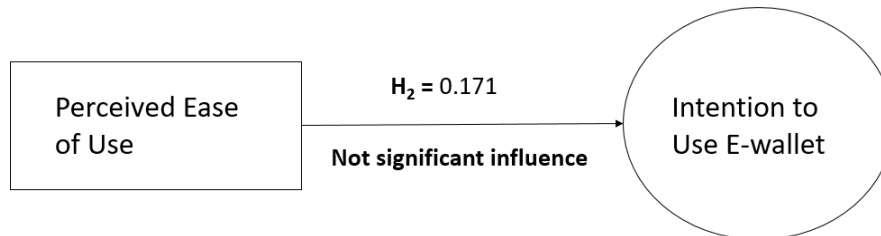


Figure 4.5 Result of Hypothesis 2

From figure 4.5, the empirical result of this study do not support H₂, in which the value of 0.171 is higher than 0.05, indicates that perceived ease of use has no significant influence on consumers' intention to use e-wallet. This result is consistent with study carried out by Yan, Md-Nor, Abu-Shannab and Sutanonpaiboon (2009) in Malaysia, where the research found that perceived ease of use is an insignificant variable to influence the consumers to adopt mobile payment. Furthermore, in study conducted by Shatskikh (2013), it agreed that effect of perceived ease of use is insignificant to influence consumer's intention to use e-wallet. According to the research done by Aslam, Ham and Arif (2017), the finding showed that perceived ease of use will not influence the consumers to adopt the e-wallet for making payment. The result of this study also agree with the conclusion made by Havidz, Aima and Wiratih (2018) that perceived ease of use does not have significant effect on consumer's intention to accept digital wallet. Additionally, the findings of this study also supported by Rahmi and Adby (2019), where people still prefer to use cash over e-wallet even though e-wallet provides lots of benefits to the users. Thus, the finding of this study is not agree with Theory of Acceptance Model (TAM) proposed by Davis (1989).

4.4.3 Research Objective 3 – Perceived Security

H₃: Perceived security has significant influence on consumers' intention to use e-wallet.

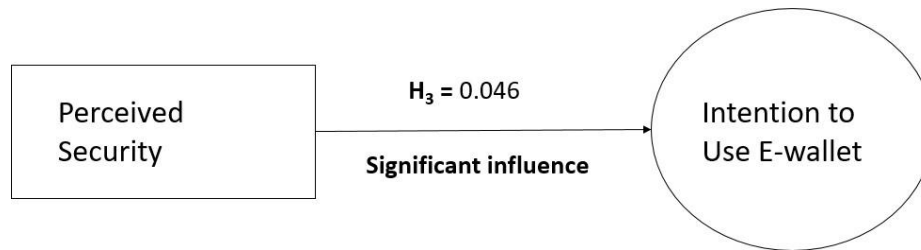


Figure 4.6 Result of Hypothesis 3

As shown in figure 4.6, the result has supported H₃, in which the value showed 0.046, lower than 0.05, indicates that perceived security is one of the significant influence that will affect consumers' intention to use e-wallet. The finding of this study is aligned with the conclusion draw in research done by Ismail (2021) that perceived security is significant indicator that has major impact on consumer's intention to use e-wallet. Security issue is the main factor that discourage people in Malaysia to adopt e-wallet. The empirical result of this study also supported by research conducted by Nizam et al., (2018) that stated that the relationship between perceived security and consumers' intention towards adopting e-wallet is significant. Humbani (2018) also found that perceived security is significant correlated with intention to adopt e-wallet, which is consistent with the result of this study. As a result, Unified Theory of Acceptance and Use of Technology (UTAUT) is proved to be reasonable as perceived security has impact on the intention of an individual to use the technology system.

4.4.4 Research Objective 4 – Subjective Norm

H₄: Subjective norm has significant influence on consumers' intention to use e-wallet.

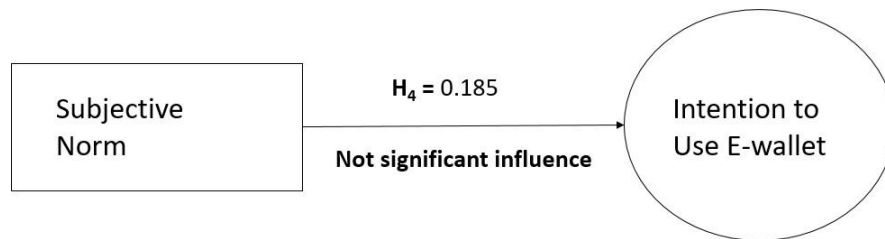


Figure 4.7 Result of Hypothesis 4

According to figure 4.7, the result of this study does not support H₄, as the value of 0.185 is greater than 0.05, indicates that there is a significant relationship between subjective norm and consumers' intention to adopt e-wallet. The result of this study is deemed to counter the Theory of Reasoned Action (TRA), as subjective norm in this study is deemed to have insignificant impact on the intention to adopt e-wallet. Finding from this study does not agree with the result of conformity experiment which was carried out by Solomon (1958), where this study concluded that social pressure will not have any impact on individual's intention to adopt e-wallet.

The result from this study is supported by research conducted by Tan, Lim and Kong (2021), in which the researchers agree that subjective norm has little relationship with the intention of adopting e-wallet. In a study done by Belousova and Chichkanov (2015), it is discovered that the social influence, which is also known as subjective norm do not have significant correlation with the intention to use electronic payment. Thus, although the result is in contrast with many past studies, but they are also studies that support this finding.

4.4.5 Research Objective 5 – Attitude

H₅: Attitude of people has significant influence on consumers' intention to use e-wallet.

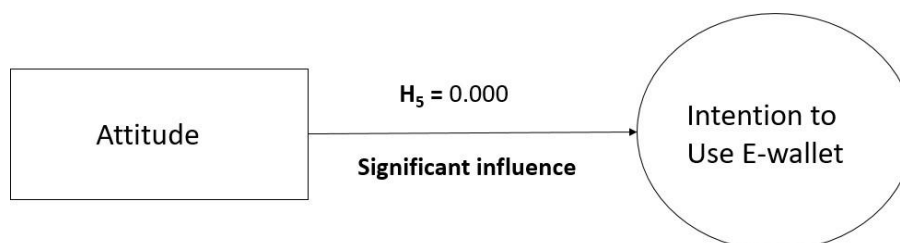


Figure 4.8 Result of Hypothesis 5

According to figure 4.8, the result showed that the attitude of an individual has significant influence on their intention to use e-wallet, which support H₅. This finding is in align with the research done by Chawla and Joshi (2019), in which it is found that the individual attitude is a significant variable to influence the individual's intention in adopting the e-wallet. Sharma, Malik and Suresh (2019) also stated that consumers' attitude is an important factor that has significant effect towards the intention to adopt e-wallet.

Based on prior studies conducted by Schierz et al., (2010), individual's attitude have positive effect on the intention to use mobile payment app and it is expected to facilitate the adoption rate. In accordance to research done by Yan, Md-Nor, Abu-Shannab and Sutanonpaiboon (2009), attitude of an individual towards using a technology system will influence the individual's behavioural intention to use the system and eventually able to predict the actual usage of the technology system. The conclusion drawn from the study by Sathish, Sermakani and Sudha (2020) is in consistent with the result of this study in which the attitude is proved to have significant influence on the consumers' intention to adopt e-wallet.

4.5 Summary of Major Findings

In this study, it is found that the relationship between perceived ease of use and intention to adopt e-wallet is not significant, unlike many studies have shown that perceived ease of use is an important determinant in influencing the intention of adoption. According to Osman and Leong (2021), there are many people who perceive that using physical cash to make payment as compared to the e-wallet, this is because new users may need time to learn the steps in

adopting the e-wallet. Furthermore, past research also mentioned that whether users have the intention to use e-wallet is much depends on the technology literacy of the individuals instead of the ease of use. The findings from this study also showed that subjective norm has no impact on consumers' intention to use e-wallet, where this is in line with research done by Lee and Wan (2010), which concluded there is no significant relationship between subjective norm and intention of e-wallet adoption.

Furthermore, the result of this study showed that perceived security is proved to have significant influence on consumers' intention to adopt e-wallet. This has proved that it is true that people in Malaysia considered security concern in adopting the e-wallet. People in Malaysia also deemed to consider the usefulness of the e-wallet, in which if the users think that e-wallet is useful to them and can make the payment process more convenient, they tend to adopt e-wallet. When potential users have established positive attitude towards using e-wallet, the adoption rate is believed to increase.

4.6 Critical Analysis

4.6.1 Managerial Implications

From this study, it has proven that perceived security has significant impact on intention to adopt e-wallet. This finding is deemed to be important for e-wallet issuer, because it implies that security concern is the most important factor to consider when developing the mobile wallet app. E-wallet issuer should take into account this findings to prioritise and address security issue in order to strengthen the security robustness of the e-wallet app. Besides that, by understanding security concern the mostly considered by the users, e-wallet can held talk or campaign to let more potential users or existing users to have better understanding about how the e-wallet app issuer has make it more secure and how users can do to strengthen their account safety. For instances, by requiring TAC verification before confirm the payment can avoid any unauthorised parties to make the payment. E-wallet providers should also develop the function of biometric authentication, though fingerprint, face recognition, or iris recognition. By implementing this function in the e-wallet app, it can enhance the security of the account and eventually boost users' confidence in adopting e-wallet.

Moreover, the findings from this study also have implication on the efforts of government in promoting people in Malaysia to adopt e-wallet. In accordance to Central Bank of Malaysia (2019), many consumers in Malaysia perceived that e-wallet has high risk of security and

privacy issues. To address the security concern of the users, the policy maker should reform the laws and regulations to inhibit the growth of cybercrime. Eventually, with robust and well-established laws and regulations, this will eventually enhance users' confidence to adopt mobile payment system and consequently lead to the achievement of a cashless society.

CHAPTER 5

5.0 Conclusion

5.1 Introduction of the Chapter

This chapter will summarise the whole research in an overview. Contribution of the study will be presented in this chapter. Besides that, limitation and recommendation for future studies will be outlined. And lastly, this chapter will end with a conclusion.

5.2 Overview of the Study

This study has discussed about the factors that have impact on the intention of consumers to adopt e-wallet. The three factors that have significant influence on the intention to use is perceived usefulness, perceived security and attitude of an individual. Whilst perceived ease of use and subjective norm do not have significant relationship with intention to use e-wallet. All the results have been analysed and supported with some past researches.

5.3 Contribution of the Study

First, e-wallet providers can obtain insights from this study, since the intention of users is being investigated and the results have showed the factors that have significant effect on driving consumers' intention to use e-wallet. E-wallet providers should pay more attention regarding the factors that have major influence on consumers' intention. To enhance the usefulness of e-wallet, the e-wallet providers can provide a timely report that analyse the consumers spending and saving, which it is usefulness for users to monitor their spending and help the users to manage their financial. Furthermore, to address the security issue or to prevent scam, e-wallet providers should build a more secure and robust system.

Second, the merchants can obtain information from this study to learn about the type of e-wallet that most of the consumers used in order to get the e-wallet payment system ready to receive payment from the customers. Besides that, the online merchants such as e-commerce retailers can benefited from this study as the findings provide insight to the retailers by understanding how important should an e-wallet system be linked to the online marketplace platform.

Third, Malaysia government can obtain insights from the findings of this study to understand the reasons that deter people to adopt e-wallet and implement laws or running campaign to increase the adoption rate in the country and eventually build a cashless society.

5.4 Limitations of the Study

It is inevitably that there are some constraints in conducting the research. Due time and method of collection constraints, the sample size that are able to collect is 245 samples, in which it may cause the result not representative of the consumers in Malaysia. Besides that, the scope of the study is narrow because most of the respondents are live Klang Valley, where it may not be representative of the population in Malaysia as this study have wide coverage of states in Malaysia.

Additionally, the age group of most of the respondents in this study is between 18 and 25 years old. This age group does not represents the age distribution of Malaysia population, as most of the responds collected are from young generation who is technology savvy and may be expert in using technology mobile devices.

5.5 Recommendation for Future Studies

For future research, quota sampling technique can be used, in order to focus on investigating how consumers with different age, gender and income level, have different perception and confidence in adopt the e-wallet. Besides that, the scope of the study shall also be expanded to cover more states in Malaysia, so as to obtain a result that is more representative of Malaysia population. After the COVID-19 pandemic ends, the future studies can be conducted to compare the results with post-pandemic and examine whether they are changes in consumers' intention to adopt e-wallet.

5.6 Conclusion

Throughout the whole research, it focuses on the factors that influence the intention of consumers to use e-wallet as the payment method in daily life. The factors that has effect on the intention to use includes perceive usefulness, perceived security, and attitude. Whereas, based on the result, perceive ease of use and subjective norm has insignificant effect on intention to use e-wallet, and the hypotheses are not supported in this study.

Nevertheless, it is undeniable that e-wallet is becoming more prevalent in today's market, and technology innovation is fast changing, hence, people should be agile in adopting these future changes.

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7.0 Appendices

Appendix A: Participant Information and Informed Consent Form

Appendix B: Survey Questionnaire

Cover letter

Title: Factors Affecting Consumers' Intention of Using E-wallet after Pandemic: A Study in Malaysia.

PLEASE READ BEFORE PROCEEDING

Good day to you.

My name is Lim Jia Le, a postgraduate student from INTI International College Subang, majoring in International Marketing under the Coventry University MBA program. I am studying about "Factors Affecting Consumers' Intention of Using E-wallet after Pandemic: A Study in Malaysia" and would like to seek your kind assistance in participating in this study. This study would entail you completing a questionnaire survey which should take no longer than 15 minutes of your time. Responses from this survey are entirely confidential and anonymous. Should you volunteer to participate, your honest answers would be highly appreciated.

Thank you very much for choosing to be a part of this study. Your cooperation will certainly contribute towards the success of this study.

Any further enquiries are welcomed. You can reach me as per the details below:

Lim Jia Le

j21034998@student.newinti.edu.my

016-3321833

Part A: Demographic Information**Instructions**

Please tick in the boxes provided for your responses.

1. Gender

☐ Male

☐ Female

2. Age group

☐ 18 – 25 years old

☐ 26 - 35 years old

☐ 36 - 45 years old

☐ 46 - 55 years old

☐ 56 years old or above

3. Race

☐ Malay

☐ Chinese

☐ Indian

☐ Others: _____ (please specify)

4. Education

- ☐ Up to high school
- ☐ Pre-university
- ☐ Diploma
- ☐ Bachelor's degree
- ☐ Master's degree
- ☐ PhD (Doctor of Philosophy)
- ☐ Others: _____ (please specify)

5. Occupation

- ☐ Student
- ☐ Worker
- ☐ Self-employed
- ☐ Retired
- ☐ Housewife
- ☐ Unemployed

6. Income / Allowance (per month)

- ☐ Below RM 1000
- ☐ RM 1000 – RM 2000
- ☐ RM 2001 – RM 3500
- ☐ RM 3501 – RM 5000
- ☐ Above RM 5000

7. Have you heard about E-wallet?☐ Yes☐ No**8. Do you use E-wallet app?**☐ Yes☐ No**9. How frequent do you use E-wallet?**☐ None☐ At least once per day☐ At least once per week☐ At least once per month**10. What type of E-wallet do you use?**☐ Alipay☐ Boost☐ Fave☐ GrabPay☐ Touch n' Go☐ ShopeePay☐ None☐ Others: _____ (please specify)

Part B**Instructions**

Please tick under the respective scales on your responses.

Scale: 1=Strongly Disagree, 2=Disagree, 3=Neutral, 4=Agree, 5=Strongly Agree

Perceived Usefulness

		1	2	3	4	5
1	E-wallet is a useful mode of payment.					
2	Using E-wallet would enhance my effectiveness in making payment.					
3	Using E-wallet can save my time.					
4	Using E-wallet allow me to pay quickly.					
5	Using E-wallet would improve my shopping experience as a consumers.					
6	Using E-wallet can improve my performance in managing personal payments.					

Perceived ease of use

		1	2	3	4	5
1	E-wallet is easy to use.					
2	Understand the steps required to use E-wallet is easy.					
3	It is easy to perform the steps required to use E-wallet.					
4	I rarely get confused when I use E-wallet.					
5	I rarely make errors when using the E-wallet.					

Perceived Security

		1	2	3	4	5
1	I feel secure with using E-wallet to make payment.					
2	I feel comfortable with having my E-wallet integrated into my mobile phone.					
3	The risk of an authorized third party overseeing the payment is low.					
4	The risk of abuse of billing information (i.e. credit/debit card number, bank account data) is low.					
5	I believe that overall riskiness of using E-wallet is low.					

Subjective Norm

		1	2	3	4	5
1	People who are important to me believe that using E-wallet is beneficial.					
2	People who are important to me would consider using E-wallet to be a good idea.					
3	People who are important to me would recommend using E-wallet.					
4	It is expected that people that are similar to me would use E-wallet.					

Attitude

		1	2	3	4	5
1	I like the idea of using E-wallet for transactions.					
2	It is a wise idea to use E-wallet for transactions.					
3	Using E-wallet for payment is beneficial.					
4	Using E-wallet is time-saving.					

Intention to use E-wallet

		1	2	3	4	5
1	I intend to use E-wallet in my daily life.					
2	If I could, I would use E-wallet as much as possible in the future.					
3	I would continue to use E-wallet in the future.					
4	I intend to recommend others to use E-wallet.					

Thank you again for participating in this survey. Please feel free to use the space below to provide any additional information or comment that you think might assist this questionnaire.

Appendix C: Research Ethics Form

**ETHICS APPLICATION FORM FOR ACADEMIC
RESEARCH****1. Applicant Details**

Full name of the student conducting the research

Lim Jia Le

University Partner

INTI International College Subang

Department

Centre for Postgraduate Studies

Supervisor

Dr. Poh Phaik See

Module code and name

INTM030CL - Dissertation

2. Project Summary

Project title

Factors Affecting Consumers' Intention of Using E-Wallet after the onset of Covid-19
Pandemic: A Study in Malaysia

Brief Research Project Summary

This research project aims to investigate the factors that affecting the consumers in Malaysia, their intention to use e-wallet after the onset of COVID-19 pandemic. The factors that will be examined consist of perceived usefulness, perceived ease of use, perceived security, subjective norm, and attitude.

Start and end dates (From -To)

7 January 2022 – 30 March 2022

Names of Co-Investigators (If applicable)

Nil

Is this project externally funded?

Nil

Does the project need to use Professional Code of Ethical Practice?

Nil

3. Project Detail

What are the aims and objectives of the project?

This research aims to investigate the factors that affect consumers' intention to use e-wallet after the onset of COVID-19 pandemic. The objectives of the project are:

- (a) To investigate the relationship between perceived usefulness and consumers' intention to use e-wallet.
- (b) To determine the relationship between perceived ease of use and consumers' intention to use e-wallet.
- (c) To measure the relationship between perceived security and consumers' intention to use e-wallet.
- (d) To analyse the impact of subjective norm on consumers' intention to adopt e-wallet.
- (e) To examine the effect of attitude on consumers' intention to adopt e-wallet.

Explain your research design

Ontology: Objectivism
 Epistemology: Positivism
 Type of Research: Quantitative
 Method of Data Collection: Online questionnaire survey
 Type of Sampling: Non-probability convenient sampling
 Theorisation: Deductive Theory Testing

Outline the principal methods you will use

Online Questionnaire Survey for Data Collection
 Structural Equation Modelling – Smart PLS

Are you proposing to use a validated scale or published research method / tool?

Yes

4. Data Analysis

Please tick the appropriate box

Questions	Yes	No
1 Does your research seek to understand, identify, analyse and/or report on information on terrorism or from terrorist organisations, require access to terrorist groups or those convicted of terrorist offences or relate to terrorism policies in other international jurisdictions?		✓

2 Does your research seek to understand, identify, analyse and/or report on information for other activities considered illegal in the UK and/or in the country you are researching in?		✓
3 Are you analysing Secondary Data?		✓
4 Are you dealing with Primary Data involving people?	✓	
5 Personal or Sensitive data:		
5a Are you dealing with personal data?		✓
5b Are you dealing with sensitive data (special category data)?		✓
6 Is the project solely desk based?	✓	
7 Will the data collection, recruitment materials or any other project documents be in any language other than English?		✓

5. Areas of Study

1 Which of the following areas are relevant to your study (please tic the appropriate box):

Travel away from home campus	
Biological samples, physical measurements/activities or substances/procedures administered to or taken from human participants	
Information gathered from human participants	✓
Biological materials including organisms and samples (excluding humans)	
Animals and their habitats, tissues, body fluids, cell lines derived from animal tissues (excluding humans)	
Hazardous substances	
Machinery & equipment	
Ionising Radiation	
UAS flights (drones)	
Environmental samples	

2 Are there any other ethical issues or risks of harm raised by the study that have not been covered by previous questions?

Yes	
No	✓

PROJECT CHECKLIST

1. DBS (Disclosure & Barring Service) formerly CRB (Criminal Records Bureau)

Does the study require DBS (Disclosure & Barring Service) checks?	No
Does the study involve direct contact by any member of the research team?	No

Does the study involve direct contact by any member of the research team?

a. children or young people under 10 years of age?	No
b. with adults who have learning difficulties, brain injury, dementia, degenerative neurological disorders?	No
c. with adults who are frail or physically disabled?	No
d. with adults who are living in residential care, social care, nursing homes, re-ablement centres, hospitals or hospices?	No
e. with adults who are in prison, remanded on bail or in custody?	No

Note: If any of the above answer is yes, that research ethics application will not be approved. This means, researcher cannot collect data from any of the above categories of people.

2. External Ethics Review

1 Will this study be submitted for ethical review to an external organisation?

No

3. Confidentiality, Security and Retention of Research Data

1 What data are you collecting/using/recording?

Factors affecting consumers' intention to use e-wallet after COVID-19 pandemic
(Online Survey)

2 Are there any reasons why you cannot guarantee the full security and confidentiality of any personal or confidential data collected for the study?

No

If "No", please provide an explanation

Data will be anonymous

- 3 Is there a significant possibility that any of your participants, and associated persons, could be directly or indirectly identified in the outputs or findings from this study?

If "Yes", please explain how and why

No

- 4 Is there a significant possibility that a specific organisation or agency or participants could have confidential information identified, as a result of the way you write up the results of the study?

No

If "Yes", please explain how and why

- 5 Will any members of the research team retain any personal or confidential data at the end of the project, other than in fully anonymised form?

No

If "Yes", please explain how and why

- 6 Will any member of the team intend to make use of any confidential information, knowledge, trade secrets obtained for any other purpose than the research project?

No

If "Yes", please explain how and why this is the case

- 7 Have you taken necessary precautions for secure data management, in accordance with data protection and [CU Policy](#)

Yes

- 8 Specify location (physical and electronic) where data will be stored

INTI International College Subang OneDrive (password protected)

- 9 Will you be responsible for destroying the data after study completion?

Yes

10 Please explain how any identifiable and anonymous data will be destroyed

It will be deleted from system after 8 months

11 Planned disposal date

31st December 2022

4. Participant Information and Informed Consent

1 Will all the participants be fully informed why the project is being conducted and what their participation will involve, prior to data collection?

Yes

2 Will every participant be asked to give written consent to participating in the study, before it begins?

Yes

3 Will all participants be fully informed about what data will be collected, and what will be done with this data during and after the study?

Yes

4 Please explain what recordings (audio, visual or both) will be made and how you will gain consent for recording participants

Nil

5 Will all participants understand that they have the right not to take part at any time, and/or withdraw themselves and their data from the study if they wish?

Yes

6 Will all participants understand that there will be no reasons required or repercussions if they withdraw and/or request to remove their data from the study?

Yes

7 Does the study involve deceiving, or covert observation of, participants?

No

8 Please give details

5. Payments to participants

- 1 Do you intend to offer participants inducements, payments or reward for taking part in your study?

No

Note: Researchers cannot offer participants inducements, payments or reward for taking part in the study.

6. Capacity to give valid consent

- 1 Do you propose to recruit any participants who are (please tick the appropriate boxes):

a Children or young people under 18 years of age?	No
b Adults who have learning difficulties, mental health conditions, brain injury, advanced dementia, degenerative neurological disorders?	No
c Adults who are physically disabled and cannot provide written and/or verbal consent	No
d Adults who are living in residential care, social care, nursing homes, residential centres, hospitals or hospices?	No
e Adults who are in prison, remanded on bail or in custody?	No
If you answered YES to any of the answers above, please explain how you will overcome any challenges to gaining valid consent	

Note: Researcher cannot collect data from any of the above categories of people.

- 2 Do you propose to recruit any participants with possible communication difficulties, including difficulties arising from limited use of knowledge of the English language?

Yes	
No	✓

- 3 Do you propose to recruit participants who may not be able to fully understand the nature of the study, the foreseen implications or cannot provide consent?

Yes	
No	✓

7. Recruiting Participants

- 1 Who are the participants?

Participant who aged 18 years old or above

- 2 How are participants being recruited? Please provide details on all methods of recruitment you intend to use

An online questionnaire survey via Google Form will be send to the participants through WhatsApp, email and will be shared through social media such as LinkedIn, Facebook and Instagram.

- 3 Do you foresee any conflict of interest?

No. Independent researcher with no vested interest in any e-wallet portal/industry

8. Online and Internet Research

- 1 Will any part of your project involve collecting data via the internet or social media? If "Yes", please explain.

Yes – The link of the survey will be sent to the respondents through email, WhatsApp and will be shared through social media.

- 2 Will you be collecting data using an online questionnaire/ survey tool? If "Yes", please explain.

Yes –Google form will be used by sending the link of the survey to respondents who is above 18 years old through email, WhatsApp, and also will be shared through social media such as LinkedIn, Facebook and Instagram.

- 3 Is there a possibility that the study will encourage children under 18 to access inappropriate websites, or correspond with people who pose risk of harm? If "Yes", please explain.

No

- 4 Will the study incur any other risks that arise specifically from the use of electronic media?

Yes	
No	√

9. Languages

- 1 Are all or some of the consent forms, information leaflets and research instruments associated with this project likely to be used in languages other than English?

Yes	
No	√

If "Yes", please specify the language[s] to be used

--

2 Are the translations being undertaken by you or a member of the research team?

Yes	
No	√

3 If "Yes", please describe the procedures used when undertaking research instrument translation (e.g. forward and back translation), clarifying strategies for ensuring the validity and reliability of the translation

--

10. Information gathered from human participants

1 Does your project involve primary data collection from human participants via questionnaires, focus groups, interviews, psychological tests, photography/videography etc.?

Yes	√
No	

Please detail the information to be collected and methods that will be used.

Online questionnaire survey to examine the factors that affect consumers' intention to use e-wallet after the onset of COVID-19 pandemic.

2 Is there the possibility of physical or psychological harm to the researcher(s) or the participants?

Yes	
No	√

3 Are any specific exclusions needed to prevent possible harm to participants (e.g. excluding people with known mental health problems)?

Yes	
No	√

4 Are any of the questionnaires or other tests being used in the research diagnostic for specific clinical conditions?

Yes	
No	√

Appendix D: Meeting Diary



M030LON DISSERTATION – Meeting Diary – Factors Affecting Consumers' Intention of Using E-Wallet: A Study in Malaysia

Student Name: Lim Jia Le (Jane)		Supervisor Name: Dr Poh Phaik See	
Week	Student Requirements	Summary of Actions Agreed	Meeting Attendance Signatures
Week 1	Review of Research Proposal: Research Topic and Design Meeting diary created	(1) Review RP- RQ and RO – to revise for clarity (2) Review research framework (4) Expand Lit Review and other sections (5) Review questionnaires-to add questions for the dependent variable – an important feature for quantitative research (6) Student to revise and send draft of finalized questionnaire for Supervisor's purview (7) Target to send out questionnaire by Week 2	Student: Jane Lim Supervisor: Dr Poh Phaik See Date: 7 Jan 2022
Week 3	Review of Submitted Literature Review and Methodology Meeting diary updated	(1) Review title to remove "continued" (2) Review questions for "subjective norms"; "attitude" and "intention to use..." (3) Discuss on how to write Chapter 3, section by section discussion (4) Review completed Chapter 2, to expand on some variables and parts of Chapter 3 – just a minor change for "sampling technique" should not be under probability (5) To start off survey after final amendments	Student: Jane Lim Supervisor: Dr Poh Phaik See Date: 21 Jan 2022
Week 5	Review of Submitted Data Collection Meeting diary updated	(1) Discuss Chapter 4 and 5 and review the planned structure of analysis (2) Review Chap Three, to include "Time Horizon", "sampling technique" (3) Student to proceed to do pilot test upon collection of data from 30 to 50 respondents (4) Understanding the various interpretation of results (5) To complete Chap 4 outline by next week (6) To start to use professional version	Student: Jane Lim Supervisor: Dr Poh Phaik See Date: 11 Feb 2022



Week 7	Review of Submitted Analysis Meeting diary updated	(1) Discuss on Chap 3, section by section, review some areas on ethics and limitations (2) Relook at RQ, ROs (3) Discuss on how to use Smart PLS to check discriminant validity – HTMT, bootstrapping, and more (4) Show sample of Chap 4, discuss on the findings, the P-values, the implications (5) Review Critical analysis, Conclusion (6) Review the possible meanings of the findings and what it implies to stakeholders (7) To prepare slides for VIVA on 18 March 2022, take note of number of slides, show sample of VIVA (8) Can rehearse at next meeting	Student: Jane Lim Supervisor: Dr Poh Phaik See Date: 25 Feb 2022
Week 9-10	Assessed Presentation of Complete Draft to include initial Conclusions Meeting diary updated	(1) Showed sample of Chap 4 and Chap 5, discussed on Critiques and Managerial Implications in Chap 4 and discuss how to interpret and conclude her findings (2) Student to send her Presentation slides to Supervisor by 15 March for VIVA presentation on 18 March 2022 (3) Discussed on ideas to present in a summarised manner (4) Student to complete whole thesis and submit via Safe-Assign before deadline on 30 March 2022	Student: Jane Lim Supervisor: Dr Poh Phaik See Date: 11 March 2022

Further Information:

The date and times of each tutorial will be allocated by Coventry University London Campus. Tutorials require compulsory attendance on campus. Students are required to submit specific items of work 4 days in advance of each tutorial. The submitted work will typically form the basis of each tutorial.

Dissertation supervision meetings are a compulsory part of the dissertation module and all students must attend their face to face dissertation meetings with their tutor. Non-attendance at the agreed tutorial time will be counted as absent, and cannot be re-arranged.

A maximum of one tutorial can be delivered virtually, typically where data collection is being carried out. This will be agreed in advance by the Supervisor. This does not apply to the following tutorials:

- Week 5 Review of Data Collection
- Week 10 Assessed Presentation of Complete Draft to include initial Conclusions

It is the responsibility of the student to:

- Provide an agenda for each tutorial 4 days in advance
- Attend on time



- Maintain a log of meetings which are signed by both Supervisor and Student
- Include completed logs in the Appendix of the final submitted dissertation